The Consumption Functions in a Tractable Heterogeneous Agent New Keynesian Model*

Gabriele Buontempo

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Abstract

This paper derives analytical consumption functions in the Tractable Heterogeneous Agent New Keynesian (THANK) model of Bilbiie (2025) at the household level. The Saver consumption function maps expected sequences of Saver income, Hand-to-Mouth income, and real interest rates, together with beginning-of-period liquid assets, into current consumption. The Hand-to-Mouth consumption function instead links consumption to current income and beginning-of-period liquid assets. The resulting expressions show how idiosyncratic transition risk and the asset-pooling structure of THANK jointly generate both self-insurance and front-loading in Saver consumption. Under standard calibrations, the Saver consumption profile tilts toward the present and the Saver impact MPC out of current income exceeds its representative-agent counterpart. The household-level consumption functions also make it possible to decompose aggregate intertemporal MPCs into constrained and unconstrained cohort contributions. Applied to the comparison with the HANK benchmark of Auclert et al. (2024), this decomposition shows that THANK reproduces both the aggregate income iMPC profile and the quantitative split of the response across household cohorts: constrained households account for most of the impact response, while unconstrained households account for most of its persistence.

Keywords: Heterogeneous agents; Hand-to-Mouth households; consumption functions; marginal propensities to consume; tractable HANK.
JEL-Codes: E21, E32, E52, D31.

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1. Introduction

The synthesis of heterogeneous-agent and New Keynesian frameworks, now commonly referred to as HANK, represents one of the major recent developments in macroeconomic modeling. Its appeal is usually attributed to its ability to succeed where its canonical representative-agent counterpart falls short: reproducing some key empirical moments estimated from large-scale microdata, such as the marginal propensities to consume out of income shocks and the distributional effects of monetary policy.

While HANK models have demonstrated remarkable progress in matching the empirical evidence, their quantitative nature and reliance on involved numerical solutions often make their underlying economic mechanisms difficult to isolate. As a result, a growing complementary literature has emerged, aiming to provide analytical insights into the HANK framework. A fundamental step in this direction is the *Tractable* HANK (THANK) model of Bilbiie (2025): a framework that retains the key features of HANK models, while at the same time allowing for fully analytical solutions.

In that context, the aim of this paper is twofold: first, to derive individual (rather than aggregate) consumption functions in THANK; and second, to show that these functions can be used to analytically characterize the household-level intertemporal marginal propensities to consume (iMPCs) and their distributional properties, which otherwise cannot be recovered from aggregate objects alone.

THANK is an ideal laboratory to study how heterogeneity at the household level shapes aggregate dynamics. While it cannot capture the richness of full heterogeneous-agent distributions, it preserves the fundamental ingredients required for this purpose: the distinction between unconstrained Savers and constrained Hand-to-Mouth (HtM) households, the presence of liquid and illiquid assets, and the emergence of a precautionary-saving motive driven by idiosyncratic uncertainty. In addition to embedding these key features, THANK calibrations are able to match empirical moments, and therefore provide a useful tool to understand their drivers. In that regard, one of the main results of Bilbiie (2025) is that THANK can match the aggregate iMPC profile estimated in the data, as does the full HANK benchmark of Auclert et al. (2024).

The primary contribution of this paper is to show that THANK reproduces not only the *aggregate* income iMPC profile of HANK, but also its *decomposition* into constrained and unconstrained household cohorts. Figure 1 reports two panels: the left one shows the HANK model of Auclert et al. (2024), and the right one shows the THANK model of Bilbiie (2025). Both models are calibrated exactly as in their respective articles, targeting the same empirical moments of the aggregate iMPC profile out of an income shock at time $t = 0$ from Fagereng et al. (2021). The fact that both calibrations are successful in matching this aggregate profile, shown by the black lines, is one of the main results of both papers. What this figure adds is that both models, once calibrated to match the aggregate, endogenously generate similar contributions from the HtM and Saver cohorts, shown by the orange and blue lines, respectively. In other words, the simple, purposefully parsimonious, analytical asset structure in THANK can reproduce key implications of the rich asset dynamics generated by HANK distributions. While the

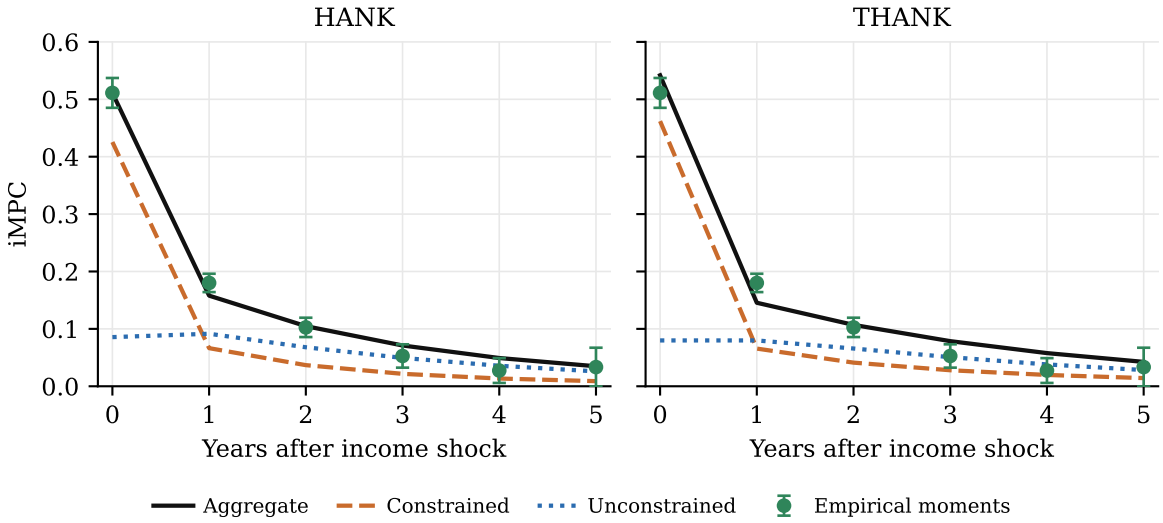


Figure 1: Fixed-cohort decomposition of the aggregate iMPC in HANK and THANK.

Notes: The HANK panel uses the HA-two benchmark of Auclert et al. (2024), with HtM households defined by the liquid-asset threshold $b \leq 0.05$. The THANK panel uses the calibration of Bilbiie (2025). Green dots report the empirical iMPC moments from Fagereng et al. (2021).

constrained HtM household cohort drives the MPC on impact of the shock, the unconstrained Saver cohort is the main driver of its persistence over periods. That this decomposition is qualitatively similar in both models is intuitive, but its quantitative correspondence across both models suggests that THANK can be used as a laboratory to study how the underlying household-level heterogeneity shapes the aggregate dynamics.

In order to produce the right panel of Figure 1, this paper derives the THANK consumption functions at the household level, separately for Savers and HtM households. These consumption functions constitute a sequence-space representation of the THANK household block: they map the initial asset position and the sequences of income and real-rate shocks into consumption responses. This is the THANK analogue of the sequence-space Jacobians of Auclert et al. (2024), from which the left panel of Figure 1 is constructed.

This paper is related to a vast and rapidly growing literature on HANK models. In particular, it contributes to the broader dialogue between quantitative richness and analytical tractability by focusing on household-level consumption functions. Quantitative HANK models discipline aggregate dynamics through rich household heterogeneity, as in Kaplan et al. (2018) and Bayer et al. (2024), among *many* others, and sequence-space methods provide a powerful way to characterize and compute these dynamics numerically (Auclert et al., 2024). A complementary analytical literature seeks to isolate the mechanisms behind these dynamics in simpler environments, building on the two-agent and hand-to-mouth tradition associated with rule-of-thumb consumers and limited asset-market participation (Bilbiie, 2008, 2020; Campbell and Mankiw, 1989; Galí et al., 2007), as well as more recent tractable HANK frameworks emphasizing monetary transmission, income risk, self-insurance, and aggregate-demand amplification (Acharya and Dogra, 2020; Bilbiie, 2025; Broer et al., 2020; Debortoli and Galí, 2024, 2025; Ravn and Sterk, 2021). The contribution of this paper is to derive analytical household-level consumption functions in THANK. I use these functions here to decompose

the aggregate iMPC path, but the same analytical representation can potentially be used to study how other components of household-level heterogeneity shape aggregate dynamics in these environments.

The remainder is organized as follows. Section 2 presents the THANK model. Section 3 derives the consumption functions in a two-period THANK model, which serves as a simplified framework for describing the main mechanisms. Section 4 extends the derivation to the infinite-horizon model and discusses the main result shown in Figure 1. Section 5 concludes. The details of the derivations are presented in Appendix A–D.

2. The THANK Model

This section describes the Tractable Heterogeneous Agent New Keynesian (THANK) framework, as presented in Bilbiie (2025) with minor notational changes and adjustments.

2.1. Savers, Hand-to-Mouth, and Asset Structure

Household States and Transitions. The economy is populated by a unit mass of households that are either unconstrained *Savers*, denoted with superscript S, or constrained *Hand-to-Mouth* (HtM), denoted with superscript H. A fraction λ of households are HtM, while the remaining fraction $1-\lambda$ are Savers. Household states evolve according to an exogenous two-state Markov chain, that generates the idiosyncratic uncertainty faced by households in this economy: conditional on being in state S in period t, a household remains a Saver in period $t+1$ with probability s and transitions to state H with probability $1-s$. Similarly, conditional on being in state H in period t, a household remains HtM with probability h and transitions to state S with probability $1-h$.

The stationary distribution of household states is determined by the invariant distribution of the Markov transition matrix. Let $(\lambda,\ 1-\lambda)$ denote the stationary fraction of households in states (H, S). These fractions satisfy

$$\begin{pmatrix} \lambda & 1-\lambda \end{pmatrix} \begin{pmatrix} h & 1-h \\ 1-s & s \end{pmatrix} = \begin{pmatrix} \lambda & 1-\lambda \end{pmatrix}.$$

Solving for the stationary distribution yields that the unconditional probability of being in the HtM state is given by

$$\frac{\lambda}{1-\lambda} = \frac{1-s}{1-h} \quad \Longleftrightarrow \quad \lambda = \frac{1-s}{2-s-h}. \tag{1}$$

This formulation nests the standard *Two-Agent New Keynesian* (TANK) framework (see Bilbiie, 2008, Bilbiie, 2020): when states are absorbing ($s = h = 1$), the fraction sizes are not pinned down by transition probabilities and λ is exogenous.

Assets and Portfolio Choice. Households can hold two types of assets: *illiquid stocks*, which represent claims on firm profits, and *liquid bonds*, which correspond to public debt. In period t, households choose current consumption and the asset portfolio they carry into period $t+1$.

Idiosyncratic uncertainty arises because households choose their asset positions in period t *before* knowing whether they will be in the Saver or HtM state in period $t+1$.

The timing of the household decision problem is as follows. At the beginning of period t, each household's current state, S or H, is revealed. Conditional on the realized state distribution, assets are pooled within each state group and redistributed equally among households in that group. Each household therefore receives the per-capita asset holdings of its current state group, together with the associated returns. Stock holdings and dividend payments are assumed to accrue only to households in the Saver state. Hence, under these pooling assumptions, the model features full insurance *within* each current state group, but incomplete insurance *across* states.

By the end of period t, households choose consumption and determine their asset holdings for period $t+1$. Savers can allocate their portfolio across both bonds and stocks, whereas HtM households can hold only bonds.

At the beginning of period $t+1$, the new states are revealed, and assets are again pooled and redistributed within each current state group. Assets available to Savers in period $t+1$ consist of the assets carried by households who were Savers in period t and remained Savers, together with the assets carried by households who were HtM in period t and became Savers. Similarly, assets available to HtM households in period $t+1$ consist of the assets carried by households who remained HtM, together with the assets carried by households who were Savers in period t and became HtM.

Notation. Stock holdings are denoted by Θ_t. Let V_t denote the market value of a stock and D_t the real dividend paid per stock. Since households that transition to state H cannot hold stocks, equity is only available to Savers. It should be noted that stocks are presented here to describe the full model, however the consumption functions derivation of the following sections will assume, for simplicity, that $\Theta_t = 0$, as these assets (unlike bonds) cannot be used for Savers' self-insurance against state-transition risk.

Bond holdings depend on household's state $J \in \{S, H\}$ and are denoted by B_t^J and Z_t^J. These variables capture the timing of portfolio decisions: Z_{t+1}^J denotes the bond holdings of a household in state J at the *end of period* t, after the consumption–saving decision has been made but *before* the household's state in period $t+1$ is revealed. Instead, B_{t+1}^J denotes the bond holdings of a household in state J at the *beginning of period* $t+1$, after the state realization has occurred. Therefore the distinction between Z_{t+1}^J and B_{t+1}^J reflects the pooling and redistribution of assets after the state transitions have happened.

Asset Redistribution Across States. Given the transition probabilities across states, the aggregate bond holdings received by each group at the beginning of period $t+1$ reflect the flows of households between the two states. Hence, from accounting, the aggregate bond holdings of the Savers group must satisfy

$$(1-\lambda)B_{t+1}^S = (1-\lambda)sZ_{t+1}^S + \lambda(1-h)Z_{t+1}^H. \tag{2}$$

Namely, $(1-\lambda)B^S_{t+1}$ are the bond holdings available to the Savers at the beginning of period $t+1$. They consist of $(1-\lambda)sZ^S_{t+1}$, the bonds carried by the Savers in period t who remained Savers, plus $\lambda(1-h)Z^H_{t+1}$, the bonds carried by the HtM in period t who became Savers. Similarly, for the Hand-to-Mouth group

$$\lambda B^H_{t+1} = (1-\lambda)(1-s)Z^S_{t+1} + \lambda h Z^H_{t+1}. \tag{3}$$

has to hold. Using the relationship implied by the stationary distribution $\lambda(1-h) = (1-\lambda)(1-s)$ in (1), the above expressions respectively simplify to

$$B^S_{t+1} = sZ^S_{t+1} + (1-s)Z^H_{t+1}; \tag{4}$$
$$B^H_{t+1} = (1-h)Z^S_{t+1} + hZ^H_{t+1}. \tag{5}$$

In sum, the bond holdings within each group at the beginning of period $t+1$ are a weighted average of the bond holdings of both states at the end of period t, with weights determined by the transition probabilities between states.

2.2. Dynamic Programming Problem

Households derive utility from consumption C_t and incur disutility from supplying labor N_t. The utility function is denoted $U(C_t, N_t)$ and future utility is discounted at the rate $\beta \in (0,1)$. Let P_t denote the price level of consumption goods at time t, and let W_t denote the real wage.

A crucial assumption of Bilbiie (2025) is that households' current states, S or H, determine to which *state-contingent group* they belong. Within each group, the consumption decisions and asset pooling are represented by a family head who assigns equal weight to the utility of each household in that group. These state-contingent groups are therefore akin to "islands", in the sense that resources are pooled and distributed within each island, while insurance across islands remains incomplete.[1]

Under standard compactness assumptions, with a discount factor $\beta \in (0,1)$, it is natural to represent the consumption problem of the family head recursively, where allocations are chosen for the state-contingent groups, assigning weights equal to the stationary population shares given by (1). The program takes the form of a Bellman equation in which the value function depends on the beginning-of-period asset holdings of each household group. Denote by $\Lambda(\cdot)$ the value function and the problem can be written as

$$\Lambda\left(B^H_t, B^S_t, \Theta_t\right) = \max_{\substack{C^H_t, C^S_t, N^H_t, N^S_t \\ Z^H_{t+1}, Z^S_{t+1}, \Theta_{t+1}}} \left\{\lambda U(C^H_t, N^H_t) + (1-\lambda)U\left(C^S_t, N^S_t\right)\right\} + \beta \mathbb{E}_t\left[\Lambda\left(B^H_{t+1}, B^S_{t+1}, \Theta_{t+1}\right)\right]$$

[1] The island analogy refers to the seminal contributions of Phelps (1970) and Lucas (1972), where the economy is populated by distinct islands. Unlike in the Lucas islands model, the role of islands here is not to generate imperfect information, but to describe state-contingent pools of resources.

subject to the constraints:

$$\begin{aligned} P_t C_t^S + Z_{t+1}^S + V_t \Theta_{t+1} &= W_t N_t^S + R_t B_t^S + (V_t + P_t D_t)\Theta_t \\ P_t C_t^H + Z_{t+1}^H &= W_t N_t^H + R_t B_t^H \\ B_{t+1}^S &= s Z_{t+1}^S + (1-s) Z_{t+1}^H \\ B_{t+1}^H &= (1-h) Z_{t+1}^S + h Z_{t+1}^H \\ Z_{t+1}^S, Z_{t+1}^H, \Theta_{t+1} &\geq 0 \end{aligned}$$

Denoting $\Pi_{t+1} \equiv \frac{P_{t+1}}{P_t}$, the resulting KKT condition for the bonds of a Saver yields:

$$\beta \mathbb{E}_t \left[\frac{R_{t+1}}{\Pi_{t+1}} \left(s U_C \left(C_{t+1}^S \right) + (1-s) U_C \left(C_{t+1}^H \right) \right) \right] = U_C \left(C_t^S \right) \quad \text{together with} \quad Z_{t+1}^S > 0 \tag{6}$$

and, focusing on the equilibria where the constraint on the Hand-to-Mouth always binds, the KKT condition for the bonds of a HtM yields:

$$\beta \mathbb{E}_t \left[\frac{R_{t+1}}{\Pi_{t+1}} \left((1-h)\, U_C \left(C_{t+1}^S \right) + h U_C \left(C_{t+1}^H \right) \right) \right] < U_C \left(C_t^H \right) \quad \text{together with} \quad Z_{t+1}^H = 0 \tag{7}$$

The complete derivation of the Euler equations is provided in Appendix A.

The Saver Euler equation (6) has a state-contingent interpretation. By purchasing one additional unit of liquid bonds at the end of period t, a Saver gives up current consumption in exchange for a real payoff R_{t+1}/Π_{t+1} next period. Since the household does not know whether they will be in state S or H in period $t+1$, this payoff is valued using the probability-weighted marginal utility across future states. Hence, $(1-s)$ measures the idiosyncratic risk faced by the household, and the term $(1-s)U_C(C_{t+1}^H, N_{t+1}^H)$ captures the precautionary saving motive: liquid bonds are valuable because they transfer resources across time and future states, in the event in which the household becomes HtM. In other words, if marginal utility in the future HtM state is high, the value of saving for a current Saver increases.

The HtM condition is instead an inequality since, in the equilibria considered here, the bond constraint binds and $Z_{t+1}^H = 0$. HtM consumption is therefore pinned down by the current budget constraint, which includes the pooled resources brought by Savers who transition into the HtM state.

Thus, the forward-looking intertemporal margin is governed by the Saver Euler equation (6), and the state-contingent consumption functions can be derived in the following sections.

3. Self-Insurance and Front-Loading in the Two-Period Model

Before turning to the infinite-horizon problem, this section derives the state-contingent consumption functions in a THANK model with only two periods. Instead of deriving the *aggregate* consumption function, I describe the consumption decisions of households in each state. These are obtained by combining the state-contingent Euler equations with the relevant budget constraints, following the household-level derivation in Luetticke et al. (2026). The two-period set-

ting is useful because it makes transparent how the risk of becoming Hand-to-Mouth changes the Saver's intertemporal margin *before* aggregation is imposed.

The key mechanism of the model is that the uncertainty induced by the exogenous transitions between the Saver and HtM states distorts intertemporal consumption. In particular, because Savers may become HtM in the future, two opposed motives arise: on the one hand there is precautionary saving, as Savers may want to self-insure against the risk of becoming constrained HtM; on the other hand, there is front-loading, as the asset structure delivers that bonds can only imperfectly finance future Saver consumption.

The aim of this section is to explicitly show how idiosyncratic transition risk and asset structure drive these two motives. Section 4 of this paper, instead, evaluates the same consumption functions in an infinite-horizon setting, which allows to assess the relative *magnitude* of these two motives, and therefore the intertemporal consumption profiles in THANK.

3.1. Derivation of the State-Contingent Consumption Functions

Here, I focus on a simplified version of the model with only two periods $t \in \{0, 1\}$ where I make the following assumptions: there are no stocks ($\Theta_t = 0$), households in the *Saver* state can only save in the form of bonds Z, and the profits from ownership of the firm are distributed in the form of dividends. Total nominal dividends of the firm are denoted $P_t D_t$ and, given that the fraction of Savers is $1 - \lambda$, each Saver receives $P_t \frac{D_t}{1-\lambda}$ at time t.

For expositional purposes, when relevant, I present each equation twice: first in the THANK setting with exogenous probabilities to transition between the two states of households (*Savers* and *HtM*), and second in the limiting case where these states are absorbing (i.e., the probabilities for a household to remain in their current state are $s = h = 1$) that serves as a comparison benchmark.

The derivations below start by manipulating the intertemporal budget constraint of a Saver and of a HtM, I then combine the two to obtain an intertemporal constraint for the aggregate bond holdings in this economy. Next, I use the log-linearized Euler equation of a Saver to derive the consumption function of a Saver, and log-linearize the budget constraint of a HtM to obtain the consumption function of a HtM.

Budget constraint of a Saver The fraction of Savers in the economy is $1-\lambda$ while the probability for a HtM to become a Saver is $1 - h$. Therefore, aggregate bond holdings received at the beginning of period $t = 1$ in the Savers group follow from the accounting identity in equation (2). Rearranging for an individual Saver household yields

$$B_1^S = \frac{1}{1-\lambda} \left[(1-\lambda) s Z_1^S + \lambda (1-h) Z_1^H \right]. \tag{8}$$

Hence the bonds distributed within the Savers group at the beginning of period $t = 1$ consist of the bonds held by Savers at the end of $t = 0$ who remain Savers at $t = 1$, and the bonds held by HtM households who transition into the Saver state. Under absorbing states ($s = h = 1$), this simplifies to $B_1^S = Z_1^S$ and the distinction between Z and B is no longer needed.

Following Bilbiie (2025), I focus on equilibria in which the borrowing constraint of the HtM households always binds. Then, $Z_0^H = 0$ and the Euler equation of the HtM (7) is a strict inequality. Under this restriction, equation (8) simplifies to

$$B_1^S = sZ_1^S. \tag{9}$$

It follows from equation (9) that, rearranging the budget constraint of a Saver in *real* notation, the bond holdings of a Saver in period $t = 0$ satisfy

$$\tilde{Z}_0^S = \frac{1}{s}\frac{1}{\tilde{R}_0}\left(C_0^S - \tilde{Y}_0^S\right) + \frac{1}{s}\frac{1}{\tilde{R}_0}\tilde{Z}_1^S \tag{10}$$

where $\tilde{Z}_t^S = \frac{Z_t^S}{P_{t-1}}$ is the household's real bond holdings, $\tilde{Y}_t^S = \frac{W_t N_t^S}{P_t} + \frac{D_t}{1-\lambda}$ is the household's real non-interest income in period t, and $\tilde{R}_t = \frac{R_t}{\Pi_t}$ is the real interest rate carried by the bonds in the beginning of period t, with $\Pi_t = \frac{P_t}{P_{t-1}}$ and $P_{-1} \equiv 1$.

Similarly, from the budget constraint, the constraint $Z_1^H = 0$ and from the transversality condition $Z_2^S = 0$, the bond holdings of a Saver in period $t = 1$, in *real* notation, satisfy

$$\tilde{Z}_1^S = \frac{1}{s}\frac{1}{\tilde{R}_1}\left(C_1^S - \tilde{Y}_1^S\right) \tag{11}$$

and combining equations (10) and (11) yields

$$s\tilde{R}_0\tilde{Z}_0^S = \left(C_0^S - \tilde{Y}_0^S\right) + \frac{1}{s}\frac{1}{\tilde{R}_1}\left(C_1^S - \tilde{Y}_1^S\right). \tag{12}$$

Thus, for a Saver, *current* bond holdings can finance *current* consumption C_0^S and *future* consumption C_1^S. However, if they are used to finance future consumption, for every C_1^S the household needs to carry $\frac{1}{s}$ units of savings in period $t = 1$, before accounting for the real return.

Given that the fraction of Savers is $1 - \lambda$, aggregating the Savers' group bonds holdings leads to

$$(1-\lambda)s\tilde{R}_0\tilde{Z}_0^S = (1-\lambda)\left[\left(C_0^S - \tilde{Y}_0^S\right) + \frac{1}{s}\frac{1}{\tilde{R}_1}\left(C_1^S - \tilde{Y}_1^S\right)\right] \tag{13}$$

which implies that only the bonds held by Savers who remain in the Saver state between periods $t = 0$ and $t = 1$ can finance current and future consumption of the Savers' group. In fact, with absorbing states ($s = 1$) this becomes

$$(1-\lambda)\tilde{R}_0\tilde{Z}_0^S = (1-\lambda)\left[\left(C_0^S - \tilde{Y}_0^S\right) + \frac{1}{\tilde{R}_1}\left(C_1^S - \tilde{Y}_1^S\right)\right] \tag{14}$$

and the current bond holdings of the Savers can freely finance their current consumption and discounted future consumption.

Budget constraint of a HtM The fraction of HtM households in the economy is λ and the probability for a Saver to become HtM is $1-s$. Aggregate bond holdings received at the beginning of period t in the HtM group are given by the accounting equation (3). Rearranging this equation, from the perspective of an individual HtM household and considering only the equilibria where the constraint on the HtM binds ($Z_t^H = 0$) yields

$$B_t^H = \frac{1}{\lambda}(1-\lambda)(1-s)Z_t^S.$$

This expression shows that the wealth available to a HtM household at the beginning of period t corresponds to the savings at the end of period $t-1$ of the Savers who became HtM in period t. If there were no Savers transitioning to the HtM state ($s = 1$), this equation would simplify to $B_t^H = 0$.

The budget constraint of a HtM in period $t = 0$ is then given by

$$P_0 C_0^H = W_0 N_0^H + R_0 \left[\frac{1}{\lambda}(1-\lambda)(1-s)Z_0^S\right].$$

Rearranging this expression in *real* notation, with $\tilde{Y}_t^H = \frac{W_t N_t^H}{P_t}$, and aggregating across the λ fraction of HtM households in this economy yields

$$(1-\lambda)(1-s)\tilde{R}_0 \tilde{Z}_0^S = \lambda\left(C_0^H - \tilde{Y}_0^H\right) \tag{15}$$

This equation shows that all consumption of the HtM in excess of their income is financed through the assets of Savers who transition into the HtM state. By contrast, with absorbing states ($s = h = 1$), this equation reads $0 = \lambda\left(C_0^H - \tilde{Y}_0^H\right)$, which corresponds to a standard setting where the HtM can only consume their current income.

Aggregating the Savers and the HtM Combining the bond holdings of the Savers (13) with the bond holdings of the HtM (15) yields an aggregate constraint for bonds that has to hold across all paths. Therefore, it has to hold in expectation, and in period $t = 0$ the aggregate bond holdings in this economy from the perspective of an individual Saver are given by

$$\tilde{R}_0 \tilde{Z}_0^S = \frac{\lambda}{1-\lambda}\left(C_0^H - \tilde{Y}_0^H\right) + \left(C_0^S - \tilde{Y}_0^S\right) + \mathbb{E}\left[\frac{1}{s}\frac{1}{\tilde{R}_1}\left(C_1^S - \tilde{Y}_1^S\right)\right]. \tag{16}$$

Hence, *current* Saver's bond holdings finance *current* consumption of the HtM as well as *current* and *future* consumption of the Savers. From this expression, it is apparent that the possibility for a Saver to become HtM acts like a tax. Part of current Savers' wealth must finance contemporaneous HtM consumption, and only a fraction of it can finance future Savers' consumption: for every C_1^S the household needs to carry $\frac{1}{s}$ units of savings into the next period. Under absorbing states ($s = 1$), this equation is simply expression (14) with both sides divided by $1-\lambda$, corresponding to a standard intertemporal consumption problem with two periods in complete markets.

Equation (16) can be log-linearized and rearranged as[2]

$$c_0^S + \frac{1}{s}\frac{1}{\tilde{R}}\mathbb{E}\left[c_1^S - \tilde{r}_1\right] = \frac{\tilde{R}\tilde{Z}^S}{C^S}\left(\tilde{r}_0 + \tilde{z}_0^S\right) - \frac{\lambda}{1-\lambda}\left(\frac{C^H}{C^S}c_0^H - \frac{\tilde{Y}^H}{C^S}\tilde{y}_0^H\right) + \frac{\tilde{Y}^S}{C^S}\tilde{y}_0^S + \frac{\frac{1}{s}\frac{1}{\tilde{R}}\tilde{Y}^S}{C^S}\mathbb{E}\left[\tilde{y}_1^S - \tilde{r}_1\right]. \tag{17}$$

This log-linearized present-value aggregate bond constraint is the starting point for deriving the consumption function of a Saver at time $t = 0$. Ultimately, the aim is to derive an expression for c_0^S on the LHS as a function of $\tilde{z}_0^S$, $\tilde{r}_0$, $\tilde{y}_0^S$, $\tilde{y}_0^H$, $\tilde{r}_1$, $\tilde{y}_1^S$, and $\tilde{y}_1^H$ on the RHS. To that end, I begin by log-linearizing the Euler equation of a Saver.

The Euler equation of a Saver Assuming that the household has CRRA preferences given by the utility function[3]

$$U(C_t, N_t) = \frac{C_t^{1-\sigma} - 1}{1-\sigma} - \nu_N \frac{N_t^{1+\psi}}{1+\psi} \tag{18}$$

the Euler equation for bonds (6) then reads

$$\beta\,\mathbb{E}\left[\tilde{R}_1\left(s\left(C_1^S\right)^{-\sigma} + (1-s)\left(C_1^H\right)^{-\sigma}\right)\right] = \left(C_0^S\right)^{-\sigma},$$

where the expected future marginal utility is an average of the marginal utility in the Saver state and the marginal utility in the HtM state, both weighted by their respective transition probabilities. That Euler equation can be log-linearized as (see Appendix B)

$$c_0^S = \mathbb{E}\left[\omega c_1^S + (1-\omega)c_1^H - \frac{1}{\sigma}\tilde{r}_1\right]. \tag{19}$$

where the weight ω is

$$\omega \equiv \frac{s}{s + (1-s)\,\zeta^\sigma} \qquad \text{with} \qquad \zeta \equiv \frac{C^S}{C^H}. \tag{20}$$

Hence, ω can be interpreted as the *share* of a Saver's expected future marginal utility derived from remaining a Saver, in steady state. It depends on ζ, the consumption ratio between the two states, raised to the power of σ: a higher curvature of marginal utility amplifies the weight of the HtM state.

An important feature of this model is that the possibility for a Saver to become HtM in the future distorts the intertemporal decision. In fact, the optimal consumption choice of a Saver takes into account that bonds can be used to self-insure against the risk of becoming HtM, and this mechanism is captured by $(1-\omega)\mathbb{E}\left[c_1^H\right]$ in the log-linearized Euler equation.

[2] All log-linearizations in this paper proceed, for a given variable X_t, by defining $X_t = Xe_t^x$ and taking the first order Taylor approximation around X, the non-stochastic steady state of X_t.

[3] Here, σ is the coefficient of relative risk aversion and hence $1/\sigma$ denotes the intertemporal elasticity of substitution. This is the reciprocal of the convention used in Bilbiie (2025).

To see how the weight ω captures the intertemporal distortion, consider the limiting case with absorbing states and therefore $s = 1$. In that case, $\omega = 1$ and optimal choice depends only on the expected future consumption of a Saver, and the expected real interest rate. This corresponds to a standard two-period model with complete markets and no heterogeneity, and the log-linearized Euler equation of a Saver (19) simplifies to $c_0^S = \mathbb{E}\left[c_1^S\right] - \frac{1}{\sigma}\mathbb{E}\left[\tilde{r}_1\right]$.

Another limiting case is the approximation around the "symmetric" steady state where the level of consumption of a Saver and a HtM are equal, i.e. $C^S = C^H \equiv C$. Then, $\omega = s$ and the log-linearized Euler equation of a Saver becomes

$$c_0^S = \mathbb{E}\left[s c_1^S + (1-s)c_1^H - \frac{1}{\sigma}\tilde{r}_1\right].$$

In this case, there is no uncertainty on the expected future steady-state consumption level and therefore $\tilde{R} = \frac{1}{\beta}$ (see Appendix B). Here, the precautionary savings motive of a Saver is only driven by the risk of transitioning into the HtM state, that is measured by $(1-s)$ the probability of this transition.

Conversely, assuming that in steady state $C^S > C^H$ implies $\omega < s$, and therefore $\tilde{R} < \frac{1}{\beta}$. Precisely, as shown in Appendix B,

$$s\tilde{R} = \frac{\omega}{\beta} \tag{21}$$

where $\frac{\omega}{s} < 1$ follows from the definition (20).

Intuitively, if the steady state consumption of a Saver is higher than that of a HtM, a Saver needs to save more in steady state to finance their expected consumption. Therefore, the possibility to become HtM in the future is more costly for a Saver compared to the limiting case $C^S = C^H$ and the weight ω on the expected future consumption of a Saver in the log-linearized Euler equation (19) is lower than the probability to remain a Saver, s. This implies that the real interest rate $\tilde{R}$ that equates the Euler equation of a Saver is lower than $\frac{1}{\beta}$ in steady state: the precautionary savings motive is amplified because of the risk of transitioning to a state where steady state consumption is lower.

The consumption functions The consumption function of a HtM at time $t = 0$ is given by log-linearizing their budget constraint (15), which yields

$$c_0^H = \frac{\tilde{Y}^H}{C^H}\tilde{y}_0^H + \frac{C^H - \tilde{Y}^H}{C^H}\left(\tilde{r}_0 + \tilde{z}_0^S\right). \tag{22}$$

Namely, current HtM consumption is a convex combination of HtM income and the bond holdings carried in period $t = 0$ by the Savers that transitioned to the HtM state. In contrast, with absorbing states ($s = h = 1$), there is no consumption of the HtM in excess of their current income: then $C^H = \tilde{Y}^H$ and this expression simplifies to $c_0^H = \tilde{y}_0^H$.

The consumption function of a Saver at time $t = 0$ is given by combining the log-linearized Euler equation of a Saver (19) and the consumption function of a HtM (22) with the log-linearized aggregate constraint for a Saver's bond holdings (17). After some algebra

described in Appendix B, rearranging yields

$$\begin{aligned} c_0^S &= \left(\frac{\omega\mathcal{F}}{\omega\mathcal{F}+\beta}\right)\frac{\tilde{Y}^S}{C^S}\left(\tilde{y}_0^S + \frac{\phi}{s\tilde{R}}\,\mathbb{E}\big[\tilde{y}_1^S\big]\right) \\ &\quad - \left(\frac{\omega\mathcal{F}}{\omega\mathcal{F}+\beta}\right)\frac{1}{s\tilde{R}}\left(\frac{\tilde{Y}^S}{C^S} + \frac{\phi-\sigma\omega}{\sigma\omega}\right)\mathbb{E}\left[\tilde{r}_1\right] \\ &\quad + \left(\frac{\omega\mathcal{F}}{\omega\mathcal{F}+\beta}\right)\frac{s\tilde{R}\tilde{Z}^S}{C^S}\left(\tilde{r}_0 + \tilde{z}_0^S\right) \\ &\quad + \left(\frac{\omega\mathcal{F}}{\omega\mathcal{F}+\beta}\right)\frac{\tilde{Y}^H}{C^H}\frac{1-\omega}{\omega}\frac{\phi}{s\tilde{R}}\mathbb{E}\left[\tilde{y}_1^H\right] \end{aligned} \tag{23}$$

where the terms ϕ and $\mathcal{F}$ are discussed in the following paragraphs. To fully grasp the consumption function (23), it is useful to consider the limiting case with absorbing states, where $C^H = \tilde{Y}^H$ and $s = h = 1$, that implies $\omega = 1$ as well as $\mathcal{F} = 1$ and $\phi = 1$, therefore

$$\begin{aligned} c_0^S &= \left(\frac{1}{1+\beta}\right)\frac{\tilde{Y}^S}{C^S}\left(\tilde{y}_0^S + \frac{1}{\tilde{R}}\mathbb{E}\left[\tilde{y}_1^S\right]\right) \\ &\quad - \left(\frac{1}{1+\beta}\right)\frac{1}{\tilde{R}}\left(\frac{\tilde{Y}^S}{C^S} + \frac{1-\sigma}{\sigma}\right)\mathbb{E}\left[\tilde{r}_1\right] \\ &\quad + \left(\frac{1}{1+\beta}\right)\frac{\tilde{R}\tilde{Z}^S}{C^S}\left(\tilde{r}_0 + \tilde{z}_0^S\right). \end{aligned} \tag{24}$$

The next subsection describes the mechanisms arising in the consumption function (23) compared to the benchmark with absorbing states (24).

3.2. The MPCs of a Saver and Intertemporal Substitution

Before interpreting (23), it is useful to understand two terms in this consumption function: ϕ and $\mathcal{F}$. Together with the Euler equation weight ω, they summarize the effects of state transitions and asset pooling on Saver consumption. Their emergence in the derivation of the consumption function is described in Appendix B.

ϕ and the state-transition feedback. The term ϕ captures the endogenous feedback arising from state transitions and asset pooling. If a Saver consumes more today and carries fewer bonds into the future, expected future HtM consumption falls. That, in turn, feeds back into current consumption through the Euler equation (19). The term ϕ appears in the derivation of (23) precisely when solving for expected future HtM consumption, $\mathbb{E}[c_1^H]$, and is defined as

$$\phi \equiv \frac{\omega\dfrac{s\tilde{R}\tilde{Z}^S}{C^S}}{\omega\dfrac{s\tilde{R}\tilde{Z}^S}{C^S} + (1-\omega)\dfrac{C^H - \tilde{Y}^H}{C^H}} \in (0,1). \tag{25}$$

The two terms in the denominator measure the consumption-financing services of Saver bonds in each future state, weighted by the corresponding weights from the Euler equation (19). The term $\frac{s\tilde{R}\tilde{Z}^S}{C^S}$ is the share of future Saver consumption financed by carried-over Saver bond

holdings. The term $\frac{C^H - \tilde{Y}^H}{C^H}$ is instead the share of future HtM consumption financed by pooled bonds.

Therefore, ϕ captures how important bonds are for the Saver's own future consumption. In particular, it measures the *share* of the bonds' consumption-financing services, as valued in the Euler equation, that accrues in the Saver's continuation state. A high ϕ means that bonds mostly remain relevant for the Saver's own future consumption. Conversely, a low ϕ means that the HtM-state feedback dominates, because pooled bonds are important for financing HtM consumption and the Euler equation weights that state heavily. With absorbing states $\omega = 1$ and $\phi = 1$: as there are no transitions, the feedback is absent.

$\mathcal{F}$ and bonds' absorption across states. The term $\mathcal{F}$ is closely related to ϕ and is defined as

$$\mathcal{F} \quad \equiv \quad \omega + (1-\omega)\frac{C^H - \tilde{Y}^H}{C^H}\frac{C^S}{s\tilde{R}\tilde{Z}^S} \quad = \quad \frac{\omega}{\phi}. \tag{26}$$

Hence, $\mathcal{F}$ measures the extent to which Saver bonds are absorbed into HtM consumption. For each *unit* of bond-financed Saver consumption (normalized to one), $\mathcal{F}$ gives the overall amount of bond-financed consumption across states, weighted by the corresponding weights of the Euler equation.

Accordingly, when

$$\frac{C^H - \tilde{Y}^H}{C^H} > \frac{s\tilde{R}\tilde{Z}^S}{C^S}, \tag{27}$$

then $\mathcal{F} > 1$, meaning that the share of bond-financed consumption is higher upon becoming HtM than upon remaining a Saver. In that case, the bonds carried by Savers serve future HtM consumption relatively *more*, and, from the perspective of a current Saver, bonds are a weaker instrument for transferring resources to the future Saver state.

Conversely, when $\mathcal{F} < 1$, bonds serve the Saver's own future relatively more. With absorbing states, there are no transitions, $\omega = 1$ and $\mathcal{F} = 1$: saving serves only the Saver's own future consumption.

Savers' MPC out of their current income: self-insurance and front-loading. In both consumption functions (23) and (24), the impact MPC out of a shock to current Saver income $\tilde{y}_0^S$ is given by the first line on the right-hand side.

As these consumption functions are in log-deviations, the level MPCs $\frac{\partial C_0^S}{\partial \tilde{Y}_0^S}$ equal the coefficient $\frac{\partial c_0^S}{\partial \tilde{y}_0^S}$ divided by the steady-state income share $\frac{\tilde{Y}^S}{C^S}$. The level MPCs are then precisely the terms $\frac{\omega\mathcal{F}}{\omega\mathcal{F}+\beta}$ in (23) and $\frac{1}{1+\beta}$ in the benchmark (24), each evaluated at its own steady state. These two terms are referred to as the *prefactors*, since they are common to all MPCs. The comparison between the two is straightforward: the direction of the inequality $\frac{\omega\mathcal{F}}{\omega\mathcal{F}+\beta} \gtrless \frac{1}{1+\beta}$ follows from the direction of the inequality

$$\omega\mathcal{F} \quad \gtrless \quad 1 \tag{28}$$

which is ex-ante ambiguous. The reason is that idiosyncratic transition risk affects the Saver's consumption through two opposing channels, which follow from the interpretation of ϕ and $\mathcal{F}$, that now deserve a full exposition: self-insurance and front-loading.

On the one hand, transition risk strengthens **self-insurance**. The Euler equation (19) assigns weight $1-\omega$ to the HtM state, in which the household is constrained. Therefore, part of the payoff from carrying bonds forward is valued in a state where the household cannot freely adjust bond holdings. Hence the value of liquid bonds rises, reducing the current Saver's consumption response to a change in resources: that is the precautionary savings motive. This motive is present whenever $s < 1$ and is amplified by steady-state inequality $\zeta > 1$, which raises the marginal utility of the HtM state and lowers ω below s. In fact, if $\mathcal{F} = 1$ (i.e. (27) holds with equality) then $\omega < 1$ implies $\omega\mathcal{F} < 1$: self-insurance alone unambiguously lowers the MPC prefactor below the absorbing-states benchmark.

On the other hand, transition and asset pooling generate **front-loading**. When $\mathcal{F} > 1$, carried bonds accrue mainly to HtM consumption. Hence, bonds transfer resources to the future Saver state only incompletely, and a Saver instead allocates a larger portion of any additional resource to current consumption. This channel operates through $\mathcal{F}$ and pushes the prefactor upward.

It is clear, however, from (28) that $\mathcal{F} > 1$ is a necessary but not sufficient condition for front-loading to prevail when comparing the impact MPCs in levels. In fact, these channels operate simultaneously, as they are two faces of the same state-transition and asset-pooling structure. That bonds are carried into the HtM state is what makes them valuable for self-insurance, since part of their payoff is received in the constrained state. At the same time, it is also what makes them less effective in transferring resources to future Saver consumption. The net effect depends on which channel dominates.

Given (28), for the MPC prefactor to be higher in THANK than in the absorbing-states benchmark, $\mathcal{F} > \frac{1}{\omega} > 1$ must hold. Since $\mathcal{F}$ itself depends on ω, this condition mixes the two channels. Substituting the definition (26) separates them, and $\omega\mathcal{F} > 1$ if and only if

$$\frac{C^H - \tilde{Y}^H}{C^H}\,\frac{C^S}{s\tilde{R}\tilde{Z}^S} \quad > \quad 1 + \frac{1}{\omega}. \tag{29}$$

The left-hand side captures the extent to which bonds are absorbed into HtM consumption, relative to Saver consumption. For the prefactor to exceed the absorbing-state benchmark, front-loading must therefore clear a hurdle $1 + \frac{1}{\omega} > 2$ that is itself increasing in the strength of self-insurance: the heavier the Euler-equation weight on the HtM state (the lower ω), the higher the amount of bonds absorbed into the HtM state must be before front-loading prevails in levels.

Savers' iMPC out of their expected future income. The first line of the consumption functions (23) and (24) also contains the Saver's consumption response to a shock to their expected future income, $\partial c_0^S / \partial \mathbb{E}\left[\tilde{y}_1^S\right]$. That is the two-period analogue of the first-row anticipation entries in the intertemporal MPC (iMPC) matrix of Auclert et al. (2024), here at the household level and for income conditional on the Saver state.

The term $\frac{\tilde{Y}^S}{C^S}$ is again dropping out of the comparison of the corresponding iMPCs in levels, as shown in the previous paragraphs. Then, as far as the prefactor is concerned, the discussion is the same as for a shock to current income of the previous paragraphs. What expected future income changes is the discount factor that multiplies it. In the benchmark with absorbing states, future income is discounted by $\frac{1}{\tilde{R}} = \beta$. In THANK, using the steady-state Euler equation (21), $s\tilde{R} = \frac{\omega}{\beta}$, the discount is

$$\frac{\phi}{s\tilde{R}} \quad = \quad \frac{\phi}{\omega}\beta \quad = \quad \frac{1}{\mathcal{F}}\beta, \tag{30}$$

and the ratio of the Saver's iMPC relative to the impact MPC is then

$$\frac{iMPC^S_{0,1}}{MPC^S_{0,0}} \quad = \quad \frac{\partial C^S_0/\partial \mathbb{E}[\tilde{Y}^S_1]}{\partial C^S_0/\partial \tilde{Y}^S_0} \quad = \quad \frac{\beta}{\mathcal{F}} \tag{31}$$

whereas in the absorbing-states benchmark, this ratio is equal to β.

Hence, equation (31) makes explicit that idiosyncratic transition-risk in THANK not only changes the level of the MPCs out of income, but also the profile of the consumption response across periods. When $\mathcal{F} > 1$, bonds are heavily absorbed into the HtM state, future income is discounted more than in the benchmark, and the consumption response *tilts*, away from future income news, and towards current income shocks.

Therefore, that front-loading is active (i.e. $\mathcal{F} > 1$) is not sufficient for the prefactor to exceed its absorbing-states benchmark in levels, but it is sufficient for the MPC profile to tilt towards current income.

The factor $\frac{1}{\mathcal{F}} = \frac{\phi}{\omega}$ is the wedge that captures the distortion induced by state transitions and asset pooling on the Saver's valuation of continuation objects: every term of (23) dated $t = 1$ is scaled by it, and no term dated $t = 0$ is. Its two components pull in opposite directions: while the state-transition feedback ϕ lowers the valuation of continuation resources, $\frac{1}{\omega}$ raises it, as the Euler equation evaluates bonds at the depressed return $s\tilde{R} = \frac{\omega}{\beta} < \frac{1}{\beta}$, and therefore the present value of future resources rises. Hence, this wedge falls below one, and therefore the consumption profile tilts towards the present, when the state-transition feedback outweighs the return effect in the Euler equation. That occurs precisely when $\mathcal{F} > 1$, and bonds serve future HtM consumption relatively more than Saver consumption.

Savers' response to the expected future real interest rate. The second line in both consumption functions (23) and (24) captures the response of current Saver consumption to the expected future real rate, $\mathbb{E}[\tilde{r}_1]$. The two coefficients share the same structure: the common prefactor, the discount, and a term between brackets combining two standard intertemporal channels: income and substitution effects.

Inside the brackets, the component $\frac{\tilde{Y}^S}{C^S}$ arises from the discounting of future income: a higher expected real rate lowers the present value of $\mathbb{E}\left[\tilde{y}^S_1\right]$. The remaining component is the standard CRRA income–substitution term. In the benchmark, that is: $\frac{1-\sigma}{\sigma} = \frac{1}{\sigma} - 1$, hence intertemporal substitution, measured by the elasticity $\frac{1}{\sigma}$, net of the income effect on the value of the consumption plan, -1.

The additional term introduced by the THANK coefficient is isolated by rewriting

$$\frac{\phi - \sigma\omega}{\sigma\omega} \quad = \quad \frac{\phi}{\sigma\omega} - 1 \quad = \quad \frac{1}{\sigma\mathcal{F}} - 1, \tag{32}$$

which is the benchmark term with the intertemporal-substitution elasticity $\frac{1}{\sigma}$ replaced by $\frac{1}{\sigma\mathcal{F}}$. Therefore, the same continuation wedge $\frac{\phi}{\omega} = \frac{1}{\mathcal{F}}$ that applies to expected future income also applies to the expected real interest rate. Intuitively, substituting consumption toward the future requires carrying resources (bonds) across periods, which only deliver the share ϕ of their financing services in the Saver's continuation state, discounted by the Euler-equation weight ω. Transition risk and asset pooling thus act on *all* forward-looking terms through the same continuation wedge: future income is discounted by $\frac{\beta}{\mathcal{F}}$ instead of β, and the effective intertemporal substitution elasticity is $\frac{1}{\sigma\mathcal{F}}$ instead of $\frac{1}{\sigma}$.

As $\mathcal{F} > 1$ signifies that bonds are absorbed relatively more in the HtM state than in the Saver state, the intertemporal substitution effect is then attenuated relative to income effect. With absorbing states, $\mathcal{F} = 1$ and the elasticity reverts to its textbook, representative agent value.

Savers' MPCs out of beginning-of-period wealth. The third line in both consumption functions (23) and (24) shows the Saver's consumption response to shocks of beginning-of-period liquid wealth, $(\tilde{r}_0 + \tilde{z}_0^S)$. This MPC applies both to shocks to inherited bond holdings and to shocks to the current return. Hence, it also captures the revaluation channel of the current interest rate.

The prefactor discussion is unchanged, and what differs is the wealth base to which the prefactor applies. In (23), that base is gross wealth income, $\tilde{R}\tilde{Z}^S$, net of $\frac{\lambda}{1-\lambda}\left(C^H - \tilde{Y}^H\right)$, which captures the amount of wealth used to finance current HtM consumption through the state-transition and asset-pooling structure, expressed per Saver. From the HtM budget constraint (15), that wealth base reduces to

$$s\tilde{R}\tilde{Z}^S \quad = \quad \tilde{R}\tilde{Z}^S - \frac{\lambda}{1-\lambda}\left(C^H - \tilde{Y}^H\right).$$

Thus, as mentioned earlier, transition risk acts like a *tax* on Saver wealth income at rate $1 - s$, the probability of leaving the Saver state. The MPC out of beginning-of-period wealth is therefore the common prefactor multiplied by $\frac{s\tilde{R}\tilde{Z}^S}{C^S}$, the same financing share that appears in the numerator of ϕ. With absorbing states, $s = 1$ and Savers retain their entire wealth income.

Savers' iMPC out of the expected future HtM income. Finally, the fourth line in (23) captures the Saver's consumption response to a shock to expected future HtM income, and has no counterpart in the benchmark (24).

This iMPC combines some terms that were described previously: the prefactor, the HtM income share $\frac{\tilde{Y}^H}{C^H}$, the ratio of Euler-equation weights $\frac{1-\omega}{\omega}$, and the discount factor $\frac{\phi}{s\tilde{R}}$ that is identical to the one applying to $\mathbb{E}\left[\tilde{y}_1^S\right]$.

The positive sign of this iMPC captures that Saver's consumption moves in the same direction as future HtM income shocks, which follows from the precautionary savings motive.

A positive future HtM income shock decreases the marginal utility from consumption in that state, which dampens self-insurance. Conversely, a negative shock raises the marginal utility of the HtM state and strengthens the precautionary motive. With absorbing states, this term disappears as the transition that generates it is absent.

Relative to expected future Saver income iMPC, the Saver's loading on expected HtM income is described by

$$\frac{\partial C_0^S / \partial \mathbb{E}[\tilde{Y}_1^H]}{\partial C_0^S / \partial \mathbb{E}[\tilde{Y}_1^S]} = \frac{(1-s)}{s} \zeta^{\sigma+1}, \tag{33}$$

which intuitively increases with transition risk $(1-s)$, risk aversion σ, and steady-state consumption inequality $\zeta \equiv \frac{C^S}{C^H}$.

The two-period model: summary and shortcomings. The two-period model clarifies how transition risk and asset structure affect the consumption profile in THANK. Two channels emerge that drive the consumption response in opposite directions: a precautionary savings motive drives self-insurance on the one hand, while the absorption of bonds into the HtM state drives front-loading on the other. Whether one channel or the other prevails is pinned down by thresholds on $\mathcal{F}$.

When $\mathcal{F} > 1$, front-loading is active: the consumption profile tilts towards the present and the intertemporal substitution effect is dampened relative to the income effect. If, in addition, front-loading clears the hurdle $\mathcal{F} > \frac{1}{\omega} > 1$, then the Saver's MPC out of their income shock increases in levels as well, compared to the benchmark with absorbing states. These features match some of the key ingredients of richer HANK frameworks, and can be described here analytically.

The natural question is then under which parameter values THANK can generate front-loading. The inequality $\mathcal{F} > 1$ can be fully characterized in terms of model primitives, s and h, and the steady-state consumption ratio $\zeta \equiv \frac{C^S}{C^H}$. In particular, $\mathcal{F} > 1$ is verified if and only if

$$(1-h)\zeta > s \tag{34}$$

as shown in Appendix B, and the parameter regions, displayed in Figure 2 of this Appendix, show that $\mathcal{F} < 1$ generally holds: for reasonable calibrations there is no front-loading in the simplified two-period THANK model, and self-insurance prevails.

This result is mostly driven by the restriction of the horizon to two periods. In fact, self-insurance arises from the Euler equation as soon as $s < 1$, regardless of how many periods follow. Front-loading, on the other hand, originates from the repeated absorption of bonds into the HtM state, as they are carried across periods. In the two-period model, there is only one possible state transition, and therefore the scope for front-loading is naturally limited.

In a nutshell: while the two-period model allows to identify the two channels in their most transparent form, it cannot be used to evaluate how their magnitudes realistically affect the consumption profiles in THANK. To some extent, the takeaway from Figure 2 is that *even* in a setting with only two periods, and under standard time-preference and risk-aversion param-

eters, THANK *can* deliver $\mathcal{F} > 1$, provided that the consumption ratio ζ and the persistence of the Saver state s are sufficiently high, and the persistence of the HtM state h is sufficiently low.

The next section of this paper shows that, in the infinite-horizon model under baseline calibrations, low liquid-asset persistence dominates the precautionary-saving motive, which flips the result relative to the two-period example. Hence, THANK then generally delivers front-loading, and a *higher* impact MPC for a Saver.

4. The Consumption Functions in the Infinite Horizon THANK

This section extends the state-contingent consumption functions of the two-period THANK model to the infinite horizon. This makes it possible to assess the relative magnitudes of self-insurance and front-loading for a Saver, and three results follow. First, the per-period wedge $1/\mathcal{F}$ of the two-period model is replaced by the stable root μ_1, which captures bonds' persistence in the Saver state. Second, under standard calibrations, the consumption profile tilts towards the present whenever the model has a solution. Third, the closed-form consumption functions yield the household-level decomposition of the aggregate iMPC, and underlie the comparison with the full HANK model in Figure 1.

4.1. Derivation

The derivation of the consumption functions in infinite-horizon THANK follows a similar intuition as the two-period model presented in Section 3, the details are shown in Appendix C. In particular, the log-linear Euler equation governing the bonds of a Saver is, as in equation (19)

$$c_t^S = \mathbb{E}_t \left[\omega c_{t+1}^S + (1-\omega) c_{t+1}^H - \frac{1}{\sigma} \tilde{r}_{t+1} \right],$$

and similarly the log-linearized flow budget constraints are

$$c_t^S = \frac{\tilde{Y}^S}{C^S} \tilde{y}_t^S + \frac{s\tilde{R}\tilde{Z}^S}{C^S} \left(\tilde{r}_t + \tilde{z}_t^S \right) - \frac{\tilde{Z}^S}{C^S} \tilde{z}_{t+1}^S, \quad \text{and} \quad c_t^H = \frac{\tilde{Y}^H}{C^H} \tilde{y}_t^H + \frac{C^H - \tilde{Y}^H}{C^H} \left(\tilde{r}_t + \tilde{z}_t^S \right)$$

for a Saver and a HtM, respectively. Combining these three equations yields an expression summarizing the dynamics of bonds:

$$\mathbb{E}_t \left[\tilde{z}_{t+2}^S \right] - \mathcal{A}_\zeta \, \tilde{z}_{t+1}^S + \beta^{-1} \tilde{z}_t^S = F_t, \tag{35}$$

where the term $\mathcal{A}_\zeta$ is a variable combining primitives s, h, β, and steady-state objects ω and ζ, while F_t is a forcing term containing incomes and interest rates. The characteristic equation of (35) is given by

$$\mu^2 - \mathcal{A}_\zeta \mu + \beta^{-1} = 0. \tag{36}$$

Its roots, denoted μ_1 and μ_2, satisfy $\mu_1 + \mu_2 = \mathcal{A}_\zeta$ as well as $\mu_1\mu_2 = \beta^{-1}$, and Appendix C shows that the saddle configuration $\mu_1 < 1 < \mu_2$ is verified if and only if

$$\frac{(1-h)\zeta}{s} > 1 - \frac{\beta}{\omega} \tag{37}$$

which is satisfied by standard calibrations of the model, where $\omega < \beta$ holds.

Factoring the equation for bond dynamics (35) using the roots μ_1 and μ_2, and imposing the transversality condition $\lim_{j\to\infty} \mu_2^{-j}\mathbb{E}_t\left[\tilde{z}^S_{t+j+1} - \mu_1\tilde{z}^S_{t+j}\right] = 0$ yields the forward solution

$$\tilde{z}^S_{t+1} = \mu_1\tilde{z}^S_t - \sum_{j=0}^{\infty} \mu_2^{-(j+1)}\mathbb{E}_t\left[F_{t+j}\right] \tag{38}$$

that clarifies the economic interpretation of the roots: $\mu_1 < 1$ captures the persistence of bonds in the Saver state, and $\mu_2 > 1$ their future discounting. A representative agent model, without idiosyncratic state-transition risk, is then nested with $\mu_1 = 1$, and $\mu_2 = \frac{1}{\beta} = \tilde{R}$.

4.2. The Consumption Functions

The consumption function of a Saver is obtained by combining the forward solution for bonds (38) together with the budget constraint of a Saver, and rearranging. The consumption function of a HtM is their log-linearized budget constraint. That leads to the following proposition.

Proposition 1 (The Consumption Functions)**.** *In the infinite-horizon THANK model, the consumption functions in period t, log-linearized around the non-stochastic steady-state, are, if the household is in the Saver state:*

$$\begin{aligned} c^S_t \quad = \quad & \left(1 - \frac{\mu_1}{\omega}\beta\right)\frac{\tilde{Y}^S}{C^S}\sum_{k=0}^{\infty}\mu_2^{-k}\mathbb{E}_t\left[\tilde{y}^S_{t+k}\right] \\ & + \left[\frac{\tilde{Z}^S}{C^S}\left(1 - \frac{\mu_1}{\omega}\beta\right)\mu_2 - \frac{1}{\omega\sigma}\right]\sum_{k=1}^{\infty}\mu_2^{-k}\mathbb{E}_t[\tilde{r}_{t+k}] \\ & + \left(1 - \frac{\mu_1}{\omega}\beta\right)\frac{s\tilde{R}\tilde{Z}^S}{C^S}\left(\tilde{r}_t + \tilde{z}^S_t\right) \\ & + \frac{\tilde{Y}^H}{C^H}\frac{1-\omega}{\omega}\sum_{k=1}^{\infty}\mu_2^{-k}\mathbb{E}_t\left[\tilde{y}^H_{t+k}\right] \end{aligned} \tag{39}$$

and if the household is in the Hand-to-Mouth state:

$$c^H_t = \frac{\tilde{Y}^H}{C^H}\tilde{y}^H_t + \frac{C^H - \tilde{Y}^H}{C^H}\left(\tilde{r}_t + \tilde{z}^S_t\right). \tag{40}$$

Proof. See Appendix C. □

The consumption function of a HtM (40) is identical to its version in the two-period model, as it follows from the same budget constraint. It also has the same interpretation: the

HtM household consumes out of their income $\tilde{y}_t^H$ and out of the beginning-of-period wealth, $\tilde{r}_t + \tilde{z}_t^S$, consisting of the bonds brought by the Savers who transition into the HtM state.

The consumption function of a Saver (39) has the same structure as the one of the two-period model: the first line captures consumption's response to changes in the sequence of the Saver real income, the second line to changes in the sequence of the real interest rate, the third line to changes in the beginning-of-period t bonds, and the fourth line to changes in the sequence of the expected HtM real income.

Savers' iMPC out of their income. From the first line of (39), for a shock that affects the Saver's income in period $t = 0$, a fraction $1 - \frac{\mu_1}{\omega}\beta$ of the additional income is consumed on impact, and the remaining fraction $\frac{\mu_1}{\omega}\beta$ is saved in the form of bonds, whose persistence in the Saver state is captured by the stable root $\mu_1 < 1$. In fact, the intertemporal MPC path can be characterized as follows.

Corollary 1 (iMPC path of a Saver in response to an income shock)**.** *For a shock that only affects the level of income of a Saver at time $t = 0$, the intertemporal MPC path in levels, resulting from the consumption function* (39)*, is given by*

$$\frac{\partial\, C_t^S}{\partial\, \tilde{Y}_0^S} \quad = \quad \frac{C^S}{\tilde{Y}^S}\frac{\partial\, c_t^S}{\partial\, \tilde{y}_0^S} \quad = \quad \left(1 - \frac{\mu_1}{\omega}\beta\right)\mu_1^t, \qquad t \geq 0. \tag{41}$$

Proof. See Appendix C. □

It follows that at time $t = 0$ the impact MPC of a Saver, compared to the representative-agent benchmark, $1 - \beta$, is higher if and only if

$$\mu_1 < \omega. \tag{42}$$

This condition has the same economic meaning as the condition for front-loading to prevail in the two-period illustration. The stable root μ_1 captures the effective persistence of bonds in the Saver state, while ω captures the valuation of remaining a Saver, from the Euler equation for bonds. When $\mu_1 < \omega$, bonds' persistence is lower than their valuation in the Saver state: saving is a weaker instrument for transferring resources to future Saver consumption, and the Saver front-loads more of the income shock into current consumption. Conversely, when $\mu_1 > \omega$, the self-insurance motive dominates and the impact response is lower than in the representative-agent benchmark. Appendix C shows that the inequality (42) can be pinned down in terms of model primitives, and that $\mu_1 < \omega$ is satisfied with standard calibrations.

To evaluate the consumption response to future income, using a notation for iMPCs akin to Auclert et al. (2024), $M_{0,k}^{S,S}$ denotes the *Saver* consumption response in period $t = 0$ to a *Saver* income shock in period $t = k$, that is

$$M_{0,k}^{S,S} \equiv \frac{\partial\, C_0^S}{\partial\, \mathbb{E}_0[\tilde{Y}_k^S]} = \left(1 - \frac{\mu_1}{\omega}\beta\right)\mu_2^{-k}.$$

That, combined with the consumption function (39), yields the following corollary result.

Corollary 2 (iMPC ratios to future income shocks)**.** *The consumption profile in the Saver state, in response to Saver income shocks at different horizons t, satisfies*

$$\frac{M^{S,S}_{0,k}}{M^{S,S}_{0,0}} = \mu_2^{-k} = \left(\beta\mu_1\right)^k . \tag{43}$$

Hence, the saddle condition (37), $\mu_1 < 1$, implies that future income in THANK is discounted more than in the representative agent model, and therefore the consumption profile tilts towards the present. In contrast with the two-period model, where under standard calibrations the tilt is absent, with infinite horizon the tilt holds whenever the model has a solution: the per-period wedge of the two-period model, $\frac{1}{\mathcal{F}} = \frac{\phi}{\omega}$, is replaced by $\mu_1 < 1$, its compounded counterpart.

Overall, the Saver's consumption response to income shocks in the infinite horizon THANK is characterized by front-loading prevailing over self-insurance: the saddle condition $\mu_1 < 1$ tilts the consumption profile towards the present, and under baseline calibration $\mu_1 < \omega$ is satisfied, so that the impact MPC to a current income shock is greater than in a representative agent framework with complete markets.

Savers' response to the expected future real interest rate. The second line of the consumption function (39) describes how current Saver consumption responds to movements in expected future real interest rates. As with future income, the sequence of expected real rates is discounted by μ_2^{-k}.

The coefficient multiplying this sequence combines two channels. The first is the intertemporal-substitution channel, captured by $-\frac{1}{\sigma\omega}$. A higher expected future real rate makes saving more attractive and tends to reduce current consumption. In a representative-agent model, the strength of this channel is governed by the intertemporal elasticity of substitution, $1/\sigma$. In THANK, the same channel is scaled by $1/\omega$, because the Euler equation weights future marginal utility across both possible states.

The second channel is a wealth-revaluation channel. If Savers hold positive liquid assets in steady state, expected future real rates also affect the value of the asset position carried through the consumption function. A higher future real rate raises the returns on the bonds that reach that future date, and this tends to increase current consumption through the beginning-of-period wealth side of the budget constraint. This channel is proportional to $\left(1 - \frac{\mu_1}{\omega}\beta\right)$, the Saver's impact MPC out of current resources: the more strongly current consumption reacts to liquid resources, the stronger the revaluation effect. Therefore, the sign of the real-rate response depends on the balance between the incentive to postpone consumption when future returns are higher and the increase in resources implied by higher returns on the liquid asset position.

This naturally leads to the next paragraph, which isolates the direct response of current consumption to beginning-of-period wealth.

Savers' MPC out of beginning-of-period wealth. The third line of the consumption function (39) captures the response to changes in beginning-of-period liquid wealth. This includes shocks to the bonds brought into the period, $\tilde{z}_t^S$, and to their real return, $\tilde{r}_t$. The MPC out of gross beginning-of-period wealth, in levels, is

$$\frac{\partial\, C_t^S}{\partial\left(\tilde{R}_t \tilde{Z}_t^S\right)} = s\left(1 - \frac{\mu_1}{\omega}\beta\right) \tag{44}$$

and its interpretation mirrors the two-period case. The factor s reflects the state-transition accounting: only the fraction s of Saver bond holdings remains in the Saver state, while the fraction $1-s$ is pooled into the HtM state. Conditional on this retained liquid wealth, this MPC is the same as the Saver's MPC out of their current non-interest income, $1 - \frac{\mu_1}{\omega}\beta$. It is lower when liquid assets are more persistent in the Saver state, μ_1 is higher, and higher when the Euler-equation valuation of the Saver state, ω, is higher.

Savers' iMPC out of HtM income. From the fourth line of the consumption function (39), the response at time $t = 0$, in levels, to news about HtM income at horizon k is given by

$$M_{0,k}^{S,H} \equiv \frac{\partial\, C_0^S}{\partial\, \mathbb{E}_0\left[\tilde{Y}_k^H\right]} \quad = \quad \frac{C^S}{C^H}\,\frac{1-\omega}{\omega}\,\mu_2^{-k} \quad = \quad \zeta^{\sigma+1}\frac{1-s}{s}\left(\beta\mu_1\right)^k, \qquad k \geq 1, \tag{45}$$

where the second equality uses the definition of ω. Intuitively, current Saver consumption only responds to news about future HtM income ($k \geq 1$). The term $\frac{1-\omega}{\omega}$ captures the relative Euler-equation weight of the HtM state compared to the Saver state. The term $\zeta = \frac{C^S}{C^H}$ rescales the response across steady-state consumption units. The term $\mu_2^{-k} = (\beta\mu_1)^k$ discounts expected HtM income at horizon k through the same forward-discounting root, $\mu_2 > 1$, that is used for future Saver income.

Future HtM income enters the Saver's consumption function through the Euler equation: a current Saver values this income because it raises consumption in the future HtM state, in which the household would be constrained. A higher expected income in a future HtM state raises future HtM consumption, and thus lowers the marginal-utility value of additional self-insurance against becoming HtM. The self-insurance motive then weakens, and current Saver consumption rises.

Therefore, comparing the Saver's consumption response at $t = 0$ to news about future HtM income, $M_{0,k}^{S,H}$, with the response to news about future Saver income, $M_{0,k}^{S,S}$, yields the following corollary.

Corollary 3 (iMPC comparison: HtM vs. Saver income shocks)**.** *For horizon $k \geq 1$, $M_{0,k}^{S,H} > M_{0,k}^{S,S}$ is verified if and only if*

$$\frac{1-s}{s}\,\zeta^{\sigma+1} > 1 - \frac{\mu_1}{\omega}\,\beta. \tag{46}$$

Since both future income sequences are discounted by the same forward-discounting term, μ_2^{-k} drops out of the comparison. The left-hand side measures how strongly current Saver

consumption responds to expected future HtM income, while the right-hand side measures the corresponding response to expected future income in the Saver state, which is equal to the Saver impact MPC out of current income. Condition (46) is satisfied at the baseline calibration: per unit of income, the Saver's consumption reacts *more* to news about HtM income than to news about Saver income.

4.3. Comparing THANK to HANK

The benchmark against which the Saver's consumption function was compared in the previous sections was an unconstrained, representative-agent framework. That comparison isolated how idiosyncratic transition risk and the asset structure of THANK affect the consumption choice of a Saver.

This section instead compares the overall THANK environment to its richer counterpart: a full HANK model. The HANK benchmark is the baseline "HA-two" model in Auclert et al. (2024). This model features two assets, one liquid and one illiquid, both subject to borrowing constraints, in the spirit of Kaplan et al. (2018). Auclert et al. (2024) show that this calibrated HANK model matches the aggregate intertemporal MPCs estimated from Norwegian administrative data in Fagereng et al. (2021).

Similarly, Bilbiie (2025) shows that a THANK economy can also be calibrated to match the same aggregate intertemporal MPC moments. What this section adds to both is the decomposition of these intertemporal MPCs into the contributions of Savers and HtM households, which are untargeted moments of the calibration.

The relevant HANK comparison: household cohorts. The THANK consumption functions of Savers (39) and HtM households (40) are contingent on the household's state in period t. Since state transitions are governed by a Markov process, these state-contingent responses can be mapped into *fixed-cohort* consumption responses. These correspond to the consumption paths of households grouped by their state *at the date of the shock*, $t = 0$, followed over time, which can be derived analytically as follows.

Corollary 4 (Cohorts contributions to the aggregate iMPC)**.** *For a shock at time $t = 0$, and denoting $\mathrm{d}c_t^S$ and $\mathrm{d}c_t^H$ the per-capita, state-contingent, consumption responses at time $t \geq 0$ respectively implied by the consumption functions of a Saver (39) and of a HtM (40), the contribution to the aggregate iMPC at time t of the cohort that is HtM at $t = 0$, denoted $C_t^{H_0}$, is given by*

$$C_t^{H_0} = \lambda \left[\Pr\left(H_t \mid H_0\right) \mathrm{d}c_t^H + \left(1 - \Pr\left(H_t \mid H_0\right)\right) \mathrm{d}c_t^S \right], \tag{47}$$

and the contribution at time t of the cohort of Savers at $t = 0$, denoted $C_t^{S_0}$, is given by

$$C_t^{S_0} = (1 - \lambda) \left[\Pr\left(H_t \mid S_0\right) \mathrm{d}c_t^H + \left(1 - \Pr\left(H_t \mid S_0\right)\right) \mathrm{d}c_t^S \right]. \tag{48}$$

Proof. See Appendix D, where $\Pr\left(H_t \mid H_0\right)$ and $\Pr\left(S_t \mid H_0\right)$ are derived analytically. □

On impact the cohort contributions correspond to the state-contingent decomposition implied by the consumption functions, and over time households within each cohort can move across states according to the THANK transition matrix. Overall, in every period, the sum of $C_t^{H_0}$ and $C_t^{S_0}$ gives the aggregate iMPC.

This fixed-cohort decomposition is the natural comparison to a HANK model, where the constrained status is endogenous. Consider an income shock in HANK at $t = 0$. In each following period, households' "state" can be classified as constrained (HtM) or unconstrained (Saver). A state-contingent decomposition by current state would then conflate two elements: the consumption response of households who were initially in that state, and the reclassification of households whose state changes endogenously because the shock moves them across the constraint. The latter has no direct counterpart in THANK, where transition probabilities are exogenous.

Decomposing the aggregate response into cohorts fixed at the date of the shock is therefore the cleaner object for comparing the two models. In both THANK and HANK, households can be classified as constrained/HtM or unconstrained/Saver at the time of the shock, and the consumption response of those initial cohorts can then be followed over periods.

The relevant HANK comparison: who are the HtM ? In THANK, the real flow budget constraint (15) of HtM households reads

$$C_t^H = \tilde{Y}_t^H + \frac{(1-\lambda)}{\lambda}(1-s)\tilde{R}_t\tilde{Z}_t^S.$$

Hence their definition is tied to their consumption behavior, rather than to their liquid asset position. A HtM household in THANK has some beginning-of-period liquidity by construction (the bonds carried by transitioning Savers), and is identified instead by the fact that it consumes all available resources by the end of the period.

By contrast, in the HANK literature, the HtM are typically defined as the households at (or close to) their liquid borrowing constraint. A definition that makes the comparison to THANK meaningful is therefore needed, and the following one is adopted for this purpose.

Definition 1 (Constrained and unconstrained households)**.** At the beginning of period t,

1. in THANK, a household in the Hand-to-Mouth state is defined as *constrained*, and a household in the Saver state is defined as *unconstrained* ;
2. in HANK, a household whose liquid asset position is below 5.5 weeks of average income is defined as *constrained*, and a household whose liquid asset position is above this threshold is defined as *unconstrained*.

The threshold of (approximately) 5.5 weeks of average income is not chosen arbitrarily. It corresponds to the first nonzero liquid-asset point of the grid on which the HANK model is solved numerically in Auclert et al. (2024). In other words, it is the lowest possible nonzero

Table 1: iMPC profiles in the data, HANK, and THANK

		HANK			THANK		
Year	Data	Aggregate	Constrained	Unconstrained	Aggregate	Constrained	Unconstrained
0	0.511	0.511	0.426	0.086	0.542	0.462	0.080
1	0.180	0.158	0.066	0.091	0.146	0.066	0.080
2	0.103	0.105	0.037	0.068	0.107	0.041	0.066
3	0.053	0.071	0.022	0.049	0.079	0.028	0.051
4	0.027	0.049	0.014	0.036	0.058	0.020	0.038
5	0.033	0.035	0.009	0.026	0.042	0.014	0.028

Notes: The table reports iMPC profiles by horizon. Constrained and unconstrained columns give the contributions of the respective household groups to the aggregate iMPC.

liquidity level that can be evaluated in HANK, while keeping its solution and calibration exactly identical to the baseline model.[4]

HANK and THANK: Aggregate Calibration and Cohorts Decomposition. Having defined households' states and cohorts, the comparison between HANK and THANK is straightforward.

Figure 1 in the introduction shows the iMPC profiles out of an income shock at time $t = 0$ in HANK and THANK. Both models are respectively calibrated following Auclert et al. (2024) and Bilbiie (2025), as reported in Table 2 in the Appendix, to match the aggregate iMPC empirical moments from Fagereng et al. (2021).[5]

That the aggregate iMPC profiles can match the empirical moments is well known, and extensively described for both models. The decomposition into Saver and HtM contributions to the aggregate is the main result of this paper.

Table 1 reports the numbers underlying the iMPC profiles shown in Figure 1. Table 2 provides the details of the calibration. The decomposition shows that both HANK and THANK attribute the largest share of the consumption response on impact at $t = 0$ to the constrained households, that is 0.426 and 0.462 respectively for each model. However, in both models, the impact MPC of the unconstrained households is also larger than what it would be in a representative agent model. The table reports contributions to the aggregate iMPC, dividing by the unconstrained cohort's exposure to the aggregate income shock gives an implied impact response at the household level of approximately 0.21 in HANK and 0.15 in THANK. Both are above the corresponding representative-agent counterpart, $1 - \beta$, equal to approximately 0.07 with the HANK-calibrated discount factor and 0.05 with the THANK-calibrated discount

[4]This threshold, which corresponds to a little more than one month of income, is also of the same order of magnitude as the liquidity thresholds commonly used to identify HtM households empirically. For example, Kaplan et al. (2014) observe average liquid balances over the pay period and, assuming uniform consumption within the period, classify households as HtM when liquid balances are no larger than one half of pay-period income.

[5]In particular, in Bilbiie (2025), THANK matches the aggregate iMPC profile with *countercyclical* inequality. That is here equivalently rendered with unequal incidence of the income shock, by adjusting the size of $\tilde{y}_0^H$ and $\tilde{y}_0^S$ to correspond to the calibrated χ, namely to satisfy $\tilde{y}_t^H = \chi \tilde{y}_t$ and $\tilde{y}_t^S = \frac{1-\lambda\chi}{1-\lambda}\tilde{y}_t$ with $\chi = 1.4 > 1$.

factor.[6] From period $t = 1$ onwards, in both models, the unconstrained households drive the largest share of the iMPC persistence over time, that are also quantitatively similar.

This comparison is informative because the cohort decomposition is not targeted by the THANK calibration. It is an implication of the household-level consumption functions derived above. The decomposition shows that, in this calibration, THANK endogenously reproduces the broad allocation of the response across households cohorts. In that sense, the analytical consumption functions make THANK useful as an analytical laboratory for studying the household-level sources of aggregate iMPC dynamics.

5. Conclusion

This paper derived analytical consumption functions in the Tractable Heterogeneous Agent New Keynesian model of Bilbiie (2025) at the household level. The derivation characterizes the consumption responses of Savers and Hand-to-Mouth households separately, as functions of the sequences of incomes and real rates and beginning-of-period liquid wealth.

The resulting expressions make transparent the mechanisms through which idiosyncratic transition risk, THANK asset-pooling, and inequality affect consumption. For Savers, transition risk generates a self-insurance motive, but the asset-pooling structure also weakens the ability of liquid bonds to transfer resources to future Saver consumption. In the infinite-horizon model, the latter channel dominates under standard calibrations. Hence, the Saver iMPC profile tilts toward the present, and the impact response exceeds its representative-agent counterpart.

The household-level consumption functions make it possible to decompose the aggregate iMPC profile into the contributions of constrained/HtM and unconstrained/Saver cohorts. Applied to the comparison with the HANK benchmark of Auclert et al. (2024), this decomposition shows that THANK reproduces not only the aggregate iMPC profile, but also the split of that response between the two groups of households. In both, constrained households account for most of the impact MPC, while unconstrained households account for most of its persistence.

These consumption functions can be useful beyond the iMPC decomposition exercise considered here. Since they map expected sequences of income and real rates into consumption, they make explicit which expectations matter for household behavior. This suggests a natural use of THANK as an analytical environment for questions involving heterogeneity in expectation formation at the household level.

[6] Direct calculation from the calibrated parameters gives $0.086/(1-0.592) = 0.21$ in HANK and $0.080/(1-0.33\times 1.4) = 0.15$ in THANK. The relevant calibration values are reported in Table 2 in the Appendix.

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Appendix A. Derivation of the Euler Equations in THANK

The problem takes the form of the Bellman equation:

$$\Lambda\left(B_t^H, B_t^S, \Theta_t\right) = \max_{\substack{C_t^H, C_t^S, N_t^H, N_t^S \\ Z_{t+1}^H, Z_{t+1}^S, \Theta_{t+1}}} \left\{\lambda U(C_t^H, N_t^H) + (1-\lambda) U\left(C_t^S, N_t^S\right)\right\} + \beta \mathbb{E}\left[\Lambda\left(B_{t+1}^H, B_{t+1}^S, \Theta_{t+1}\right)\right]$$

subject to the constraints:

$$\begin{aligned} P_t C_t^S + Z_{t+1}^S + V_t \Theta_{t+1} &= W_t N_t^S + R_t B_t^S + (V_t + P_t D_t)\Theta_t \\ P_t C_t^H + Z_{t+1}^H &= W_t N_t^H + R_t B_t^H \\ B_{t+1}^S &= s Z_{t+1}^S + (1-s) Z_{t+1}^H \\ B_{t+1}^H &= (1-h) Z_{t+1}^S + h Z_{t+1}^H \\ Z_{t+1}^S, Z_{t+1}^H, \Theta_{t+1} &\geq 0 \end{aligned}$$

This can be rewritten more succinctly, by substituting the different B assets as:

$$\Lambda\left((1-h)Z_t^S + hZ_t^H, sZ_t^S + (1-s)Z_t^H, \Theta_t\right) = \max_{\substack{C_t^H, C_t^S, N_t^H, N_t^S \\ Z_{t+1}^H, Z_{t+1}^S, \Theta_{t+1}}} \left\{\lambda U(C_t^H, N_t^H) + (1-\lambda) U\left(C_t^S, N_t^S\right)\right\} + \beta \mathbb{E}\left[\Lambda\left((1-h)Z_{t+1}^S + hZ_{t+1}^H, sZ_{t+1}^S + (1-s)Z_{t+1}^H, \Theta_{t+1}\right)\right]$$

subject to the constraints:

$$\begin{aligned} P_t C_t^S + Z_{t+1}^S + V_t \Theta_{t+1} &= W_t N_t^S + R_t\left(sZ_t^S + (1-s)Z_t^H\right) + (V_t + P_t D_t)\Theta_t \\ P_t C_t^H + Z_{t+1}^H &= W_t N_t^H + R_t\left((1-h)Z_t^S + hZ_t^H\right) \\ Z_{t+1}^S, Z_{t+1}^H, \Theta_{t+1} &\geq 0 \end{aligned}$$

The dynamic programming problem can be written as the following Lagrangian:

$$\begin{aligned} &\Lambda\left((1-h)Z_t^S + hZ_t^H, sZ_t^S + (1-s)Z_t^H, \Theta_t\right) \\ &\quad = \max_{\substack{C_t^H, C_t^S, N_t^H, N_t^S \\ Z_{t+1}^H, Z_{t+1}^S, \Theta_{t+1}}} \left\{\lambda U(C_t^H, N_t^H) + (1-\lambda) U\left(C_t^S, N_t^S\right)\right\} \\ &\quad + \beta \mathbb{E}\left[\Lambda\left((1-h)Z_{t+1}^S + hZ_{t+1}^H, sZ_{t+1}^S + (1-s)Z_{t+1}^H, \Theta_{t+1}\right)\right] \\ &\quad + \ell_t^S\left[W_t N_t^S + R_t\left(sZ_t^S + (1-s)Z_t^H\right) + (V_t + P_t D_t)\Theta_t - P_t C_t^S - Z_{t+1}^S - V_t \Theta_{t+1}\right] \\ &\quad + \ell_t^H\left[W_t N_t^H + R_t\left((1-h)Z_t^S + hZ_t^H\right) - P_t C_t^H - Z_{t+1}^H\right] \\ &\quad + m_t^S Z_{t+1}^S + m_t^H Z_{t+1}^H + n_t \Theta_{t+1} \end{aligned}$$

where ℓ_t^S, ℓ_t^H, m_t^S, m_t^H, and n_t are Lagrangian multipliers. The corresponding KKT conditions are then, with complementary slackness (omitted):[7]

$$\frac{\partial \Lambda(\cdot)}{\partial C_t^H} = 0 \iff \lambda U_C(C_t^H, N_t^H) - \ell_t^H P_t = 0$$
$$\frac{\lambda U_C(C_t^H, N_t^H)}{P_t} = \ell_t^H$$
$$\frac{\partial \Lambda(\cdot)}{\partial C_t^S} = 0 \iff (1-\lambda) U_C(C_t^S, N_t^S) - \ell_t^S P_t = 0$$
$$\frac{(1-\lambda) U_C(C_t^S, N_t^S)}{P_t} = \ell_t^S$$
$$\frac{\partial \Lambda(\cdot)}{\partial N_t^H} = 0 \iff \lambda U_N(C_t^H, N_t^H) + \ell_t^H W_t = 0$$
$$\frac{-\lambda U_N(C_t^H, N_t^H)}{W_t} = \ell_t^H$$
$$\frac{\partial \Lambda(\cdot)}{\partial N_t^S} = 0 \iff (1-\lambda) U_N(C_t^S, N_t^S) + \ell_t^S W_t = 0$$
$$\frac{-(1-\lambda) U_N(C_t^S, N_t^S)}{W_t} = \ell_t^S$$
$$\frac{\partial \Lambda(\cdot)}{\partial Z_{t+1}^H} = 0 \iff \beta h \mathbb{E}[\Lambda_{B^H}(\cdot_{t+1})] + \beta(1-s)\mathbb{E}[\Lambda_{B^S}(\cdot_{t+1})] - \ell_t^H + m_t^H = 0$$
$$\beta \mathbb{E}[h\Lambda_{B^H}(\cdot_{t+1}) + (1-s)\Lambda_{B^S}(\cdot_{t+1})] - \ell_t^H + m_t^H = 0$$
$$\frac{\partial \Lambda(\cdot)}{\partial Z_{t+1}^S} = 0 \iff \beta(1-h)\mathbb{E}[\Lambda_{B^H}(\cdot_{t+1})] + \beta s \mathbb{E}[\Lambda_{B^S}(\cdot_{t+1})] - \ell_t^S + m_t^S = 0$$
$$\beta \mathbb{E}[(1-h)\Lambda_{B^H}(\cdot_{t+1}) + s\Lambda_{B^S}(\cdot_{t+1})] - \ell_t^S + m_t^S = 0$$
$$\frac{\partial \Lambda(\cdot)}{\partial \Theta_{t+1}} = 0 \iff \beta \mathbb{E}[\Lambda_\Theta(\cdot_{t+1})] - \ell_t^S V_t + n_t = 0$$

and the Envelope Conditions are:

$$\frac{\partial \Lambda(\cdot)}{\partial Z_t^H} = 0 \iff h\Lambda_{B^H}(\cdot_t) + (1-s)\Lambda_{B^S}(\cdot_t) = \ell_t^S R_t(1-s) + \ell_t^H R_t h$$
$$\text{iterated:} \quad h\Lambda_{B^H}(\cdot_{t+1}) + (1-s)\Lambda_{B^S}(\cdot_{t+1}) = \ell_{t+1}^S R_{t+1}(1-s) + \ell_{t+1}^H R_{t+1} h$$
$$\frac{\partial \Lambda(\cdot)}{\partial Z_t^S} = 0 \iff (1-h)\Lambda_{B^H}(\cdot_t) + s\Lambda_{B^S}(\cdot_t) = \ell_t^S R_t s + \ell_t^H R_t(1-h)$$
$$\text{iterated:} \quad (1-h)\Lambda_{B^H}(\cdot_{t+1}) + s\Lambda_{B^S}(\cdot_{t+1}) = \ell_{t+1}^S R_{t+1} s + \ell_{t+1}^H R_{t+1}(1-h)$$
$$\frac{\partial \Lambda(\cdot)}{\partial \Theta_t} = 0 \iff \Lambda_\Theta(\cdot_t) = \ell_t^S(V_t + P_t D_t)$$
$$\text{iterated:} \quad \Lambda_\Theta(\cdot_{t+1}) = \ell_{t+1}^S(V_{t+1} + P_{t+1} D_{t+1})$$

[7] The following notation is used consistently: for a function $F(X, Y)$, the partial derivative $F_X \equiv \frac{\partial F(X,Y)}{\partial X}$.

Hence, denoting $\Pi_{t+1} \equiv \frac{P_{t+1}}{P_t}$, the Euler equations are:

$$\begin{aligned}
\beta\mathbb{E}[(1-h)\Lambda_{B^H}(\cdot_{t+1}) + s\Lambda_{B^S}(\cdot_{t+1})] - \ell_t^S + m_t^S &= 0 \\
\beta\mathbb{E}\left[\ell_{t+1}^S R_{t+1} s + \ell_{t+1}^H R_{t+1}(1-h)\right] &\leq \ell_t^S \\
\beta\mathbb{E}\left[\frac{(1-\lambda)U_C(C_{t+1}^S, N_{t+1}^S)}{P_{t+1}} R_{t+1} s + \frac{\lambda U_C(C_{t+1}^H, N_{t+1}^H)}{P_{t+1}} R_{t+1}(1-h)\right] &\leq \frac{(1-\lambda)U_C(C_t^S, N_t^S)}{P_t} \\
\beta\mathbb{E}\left[\frac{R_{t+1}}{\Pi_{t+1}}\left(sU_C(C_{t+1}^S, N_{t+1}^S) + \frac{\lambda}{1-\lambda}(1-h)U_C(C_{t+1}^H, N_{t+1}^H)\right)\right] &\leq U_C(C_t^S, N_t^S) \\
\beta\mathbb{E}\left[\frac{R_{t+1}}{\Pi_{t+1}}\left(sU_C(C_{t+1}^S, N_{t+1}^S) + (1-s)U_C(C_{t+1}^H, N_{t+1}^H)\right)\right] &\leq U_C(C_t^S, N_t^S) \qquad \text{(A.1)}
\end{aligned}$$

together with

$$0 = Z_{t+1}^S \left\{U_C(C_t^S, N_t^S) - \beta\mathbb{E}\left[\frac{R_{t+1}}{\Pi_{t+1}}\left(sU_C(C_{t+1}^S, N_{t+1}^S) + (1-s)U_C(C_{t+1}^H, N_{t+1}^H)\right)\right]\right\} \qquad \text{(A.2)}$$

and

$$\begin{aligned}
\beta\mathbb{E}[h\Lambda_{B^H}(\cdot_{t+1}) + (1-s)\Lambda_{B^S}(\cdot_{t+1})] - \ell_t^H + m_t^H &= 0 \\
\beta\mathbb{E}\left[\ell_{t+1}^S R_{t+1}(1-s) + \ell_{t+1}^H R_{t+1} h\right] &\leq \ell_t^H \\
\beta\mathbb{E}\left[\frac{(1-\lambda)U_C(C_{t+1}^S, N_{t+1}^S)}{P_{t+1}} R_{t+1}(1-s) + \frac{\lambda U_C(C_{t+1}^H, N_{t+1}^H)}{P_{t+1}} R_{t+1} h\right] &\leq \frac{\lambda U_C(C_t^H, N_t^H)}{P_t} \\
\beta\mathbb{E}\left[\frac{R_{t+1}}{\Pi_{t+1}}\left(\frac{1-\lambda}{\lambda}(1-s)U_C(C_{t+1}^S, N_{t+1}^S) + hU_C(C_{t+1}^H, N_{t+1}^H)\right)\right] &\leq U_C(C_t^H, N_t^H) \\
\beta\mathbb{E}\left[\frac{R_{t+1}}{\Pi_{t+1}}\left((1-h)U_C(C_{t+1}^S, N_{t+1}^S) + hU_C(C_{t+1}^H, N_{t+1}^H)\right)\right] &\leq U_C(C_t^H, N_t^H) \qquad \text{(A.3)}
\end{aligned}$$

together with

$$0 = Z_{t+1}^H \left\{U_C(C_t^H, N_t^H) - \beta\mathbb{E}\left[\frac{R_{t+1}}{\Pi_{t+1}}\left((1-h)U_C(C_{t+1}^S, N_{t+1}^S) + hU_C(C_{t+1}^H, N_{t+1}^H)\right)\right]\right\} \qquad \text{(A.4)}$$

and

$$\begin{aligned}
\beta\mathbb{E}[\Lambda_\Theta(\cdot_{t+1})] - \ell_t^S V_t + n_t &= 0 \\
\beta\mathbb{E}\left[\ell_{t+1}^S(V_{t+1} + P_{t+1}D_{t+1})\right] &\leq \ell_t^S V_t \\
\beta\mathbb{E}\left[\frac{(1-\lambda)U_C(C_{t+1}^S, N_{t+1}^S)}{P_{t+1}}(V_{t+1} + P_{t+1}D_{t+1})\right] &\leq \frac{(1-\lambda)U_C(C_t^S, N_t^S)}{P_t} V_t \\
\beta\mathbb{E}\left[U_C(C_{t+1}^S, N_{t+1}^S)\frac{P_t}{P_{t+1}}\frac{V_{t+1} + P_{t+1}D_{t+1}}{V_t}\right] &\leq U_C(C_t^S, N_t^S) \qquad \text{(A.5)}
\end{aligned}$$

together with

$$0 = \Theta_{t+1}\left\{U_C(C_t^S, N_t^S) - \beta\mathbb{E}\left[U_C(C_{t+1}^S, N_{t+1}^S)\frac{P_t}{P_{t+1}}\frac{V_{t+1} + P_{t+1}D_{t+1}}{V_t}\right]\right\} \qquad \text{(A.6)}$$

and at time t it also has to be that

$$U_C(C_t^H, N_t^H) = -U_N(C_t^H, N_t^H)\frac{P_t}{W_t} \tag{A.7}$$

as well as

$$U_C(C_t^S, N_t^S) = -U_N(C_t^S, N_t^S)\frac{P_t}{W_t} \tag{A.8}$$

Appendix B. Consumption Functions in the Two-Period THANK Model

This appendix shows the derivation of the log-linearized consumption functions of the two-period THANK model of Section 3: the consumption function of an HtM household, the consumption function of a Saver, and its counterpart in the benchmark with absorbing states. It also collects the auxiliary results used in the interpretation of these consumption functions: the steady-state return, the wealth base, the level MPCs, the discount on forward-looking terms, and the characterization of the thresholds on $\mathcal{F}$ in terms of model primitives.

B.1. Setup

There are two periods, $t \in \{0, 1\}$. Households can be in two states: *Savers* (fraction $1 - \lambda$), who can save in one-period bonds Z^S, and *HtM* households (fraction λ), whose borrowing constraint binds so that $Z_t^H = 0$ in every period. At the beginning of each period, a Saver remains a Saver with probability s and becomes HtM with probability $1 - s$; an HtM household remains HtM with probability h and becomes a Saver with probability $1 - h$. The population shares are stationary, which requires

$$(1 - \lambda)(1 - s) \;=\; \lambda\,(1 - h). \tag{B.1}$$

Bonds are pooled within each state group: households that transition carry their bond holdings into their new group, where bond holdings are shared equally among households.

Households have CRRA preferences over consumption,

$$U(C_t, N_t) \;=\; \frac{C_t^{1-\sigma} - 1}{1 - \sigma} - \nu_N\,\frac{N_t^{1+\psi}}{1 + \psi}, \tag{B.2}$$

where σ is the coefficient of relative risk aversion, so that $1/\sigma$ is the intertemporal elasticity of substitution. Only the marginal utility of consumption enters the derivation below.

Notation follows the main text: $\tilde{Z}_t^S = \frac{Z_t^S}{P_{t-1}}$ denotes real bond holdings, $\tilde{Y}_t^i$ real non-interest income of type $i \in \{S, H\}$, and $\tilde{R}_t = \frac{R_t}{\Pi_t}$ the gross real return carried by bonds at the beginning of period t. Steady-state values carry no time subscript; lower-case letters denote log-deviations from the steady state, obtained for a variable X_t by writing $X_t = Xe^{x_t}$ and taking a first-order

approximation around X. Two steady-state objects recur throughout the derivation:

$$\zeta \equiv \frac{C^S}{C^H}, \tag{B.3}$$

the steady-state consumption ratio between the two states, and the Euler-equation weight ω defined in Section B.3 below.

B.2. Budget Constraints

The Saver

The bonds available to the Saver group at the beginning of period $t = 1$ consist of the bonds of Savers who remain Savers and the bonds of HtM households who become Savers. Then, per Saver, the accounting identity (2) yields

$$B_1^S = \frac{1}{1-\lambda}\left[(1-\lambda)\, s\, Z_1^S + \lambda\,(1-h)\, Z_1^H\right] = s\, Z_1^S, \tag{B.4}$$

where the second equality uses the binding constraint of the HtM, $Z_1^H = 0$. The Saver group therefore receives only the fraction s of the previous Saver-group bond holdings: the bonds of the transitioning fraction $1 - s$ leave with them, and the incoming ex-HtM members arrive with none.

Using (B.4), the flow budget constraint of a Saver in real terms gives the bond holdings at $t = 0$ and $t = 1$ as

$$\tilde{Z}_0^S = \frac{1}{s}\frac{1}{\tilde{R}_0}\left(C_0^S - \tilde{Y}_0^S\right) + \frac{1}{s}\frac{1}{\tilde{R}_0}\,\tilde{Z}_1^S, \tag{B.5}$$

$$\tilde{Z}_1^S = \frac{1}{s}\frac{1}{\tilde{R}_1}\left(C_1^S - \tilde{Y}_1^S\right), \tag{B.6}$$

where (B.6) imposes the transversality condition $\tilde{Z}_2^S = 0$. Combining the two,

$$s\tilde{R}_0\,\tilde{Z}_0^S = \left(C_0^S - \tilde{Y}_0^S\right) + \frac{1}{s}\frac{1}{\tilde{R}_1}\left(C_1^S - \tilde{Y}_1^S\right). \tag{B.7}$$

Current bond holdings finance current and future Saver consumption, but each unit of future consumption requires carrying $\frac{1}{s}$ units of savings.

The HtM

The bonds available to the HtM group at the beginning of period t consist of the bonds of HtM households who remain HtM and the bonds of Savers who become HtM. Per HtM household, the accounting identity (3) reads

$$B_t^H = \frac{1}{\lambda}\left[\lambda\, h\, Z_t^H + (1-\lambda)(1-s)\, Z_t^S\right] = \frac{(1-\lambda)(1-s)}{\lambda}\, Z_t^S, \tag{B.8}$$

where the second equality uses the binding borrowing constraint $Z_t^H = 0$: all the wealth available to an HtM household is the savings carried in by the Savers who transitioned into the

state. The period budget constraint of an HtM household at $t = 0$ then reads, in nominal terms,

$$P_0\, C_0^H \;=\; W_0 N_0^H + R_0 \left[\frac{(1-\lambda)(1-s)}{\lambda}\, Z_0^S \right],$$

and, in real terms with $\tilde{Y}_0^H = \frac{W_0 N_0^H}{P_0}$, aggregating across the λ HtM households:

$$(1-\lambda)(1-s)\, \tilde{R}_0\, \tilde{Z}_0^S \;=\; \lambda \left(C_0^H - \tilde{Y}_0^H \right), \tag{B.9}$$

and identically at $t = 1$. All HtM consumption in excess of HtM income is financed by the bonds brought in by transitioning Savers. With absorbing states ($s = 1$), the left-hand side vanishes and $C_0^H = \tilde{Y}_0^H$: the HtM consume their income.

Aggregation

Combining (B.7) with (B.9), and noting that the resulting constraint must hold in expectation over the period-1 states, the aggregate bond constraint from the perspective of an individual Saver reads

$$\tilde{R}_0\, \tilde{Z}_0^S \;=\; \frac{\lambda}{1-\lambda} \left(C_0^H - \tilde{Y}_0^H \right) + \left(C_0^S - \tilde{Y}_0^S \right) + \mathbb{E}\left[\frac{1}{s} \frac{1}{\tilde{R}_1} \left(C_1^S - \tilde{Y}_1^S \right) \right]. \tag{B.10}$$

To log-linearize (B.10), write each variable as its steady-state value times an exponential deviation ($X_t = X e^{x_t}$) and expand to first order ($e^x \approx 1 + x$), term by term:

$$\begin{aligned}
\tilde{R}_0 \tilde{Z}_0^S \;&\approx\; \tilde{R} \tilde{Z}^S \left(1 + \tilde{r}_0 + \tilde{z}_0^S\right), \\
\frac{\lambda}{1-\lambda} \left(C_0^H - \tilde{Y}_0^H \right) \;&\approx\; \frac{\lambda}{1-\lambda} \left[C^H \left(1 + c_0^H\right) - \tilde{Y}^H \left(1 + \tilde{y}_0^H\right) \right], \\
\mathbb{E}\left[\frac{1}{s \tilde{R}_1} \left(C_1^S - \tilde{Y}_1^S \right) \right] \;&\approx\; \frac{1}{s \tilde{R}}\, \mathbb{E}\left[C^S \left(1 + c_1^S - \tilde{r}_1\right) - \tilde{Y}^S \left(1 + \tilde{y}_1^S - \tilde{r}_1\right) \right],
\end{aligned}$$

where the last line uses $\frac{1}{\tilde{R}_1} = \frac{1}{\tilde{R}} e^{-\tilde{r}_1} \approx \frac{1}{\tilde{R}}(1 - \tilde{r}_1)$ and drops second-order cross products. Substituting into (B.10), the steady-state terms cancel by the steady-state version of the same constraint, and dividing through by C^S and rearranging:

$$\begin{aligned}
c_0^S + \frac{1}{s\tilde{R}}\, \mathbb{E}\left[c_1^S - \tilde{r}_1 \right] \;=\; \frac{\tilde{R} \tilde{Z}^S}{C^S} \left(\tilde{r}_0 + \tilde{z}_0^S \right) - \frac{\lambda}{1-\lambda} \left(\frac{C^H}{C^S}\, c_0^H - \frac{\tilde{Y}^H}{C^S}\, \tilde{y}_0^H \right) \\
+ \frac{\tilde{Y}^S}{C^S}\, \tilde{y}_0^S + \frac{\tilde{Y}^S}{s \tilde{R}\, C^S}\, \mathbb{E}\left[\tilde{y}_1^S - \tilde{r}_1 \right].
\end{aligned} \tag{B.11}$$

This log-linearized present-value constraint is the backbone of the derivation: the consumption function of a Saver is obtained by eliminating c_0^H, $\mathbb{E}\left[c_1^S\right]$ and $\mathbb{E}\left[c_1^H\right]$ from it.

B.3. The Euler Equation of a Saver

With CRRA preferences, the Euler equation for bonds of a Saver reads

$$\beta\,\mathbb{E}\Big[\tilde{R}_1\left(s\left(C_1^S\right)^{-\sigma}+(1-s)\left(C_1^H\right)^{-\sigma}\right)\Big] \;=\; \left(C_0^S\right)^{-\sigma} \tag{B.12}$$

expected future marginal utility averages the two states, weighted by the transition probabilities. Define

$$\Omega \;\equiv\; s\left(C^S\right)^{-\sigma}+(1-s)\left(C^H\right)^{-\sigma}, \qquad \omega \;\equiv\; \frac{s\left(C^S\right)^{-\sigma}}{\Omega} \;=\; \frac{s}{s+(1-s)\,\zeta^{\sigma}}. \tag{B.13}$$

Then ω is the share of the Saver's expected future marginal utility attributable to remaining a Saver: the transition probability s, adjusted by the relative marginal utility of the two states, $\zeta^{\sigma}=\left(\frac{C^S}{C^H}\right)^{\sigma}$.

To log-linearize (B.12), write $C_1^S=C^Se^{c_1^S}$, $C_1^H=C^He^{c_1^H}$, $C_0^S=C^Se^{c_0^S}$ and $\tilde{R}_1=\tilde{R}e^{\tilde{r}_1}$, and expand to first order ($e^{-\sigma c}\approx 1-\sigma c$, $e^{\tilde{r}}\approx 1+\tilde{r}$):

$$\beta\tilde{R}\,\mathbb{E}\Big[(1+\tilde{r}_1)\left(s\big(C^S\big)^{-\sigma}\big(1-\sigma c_1^S\big)+(1-s)\big(C^H\big)^{-\sigma}\big(1-\sigma c_1^H\big)\right)\Big] \;=\; \big(C^S\big)^{-\sigma}\big(1-\sigma c_0^S\big).$$

Using the definitions in (B.13), the inner bracket equals $\Omega\left[1-\sigma\omega\,c_1^S-\sigma(1-\omega)\,c_1^H\right]$, so that, dropping second-order cross products,

$$\beta\tilde{R}\,\Omega\,\mathbb{E}\big[1+\tilde{r}_1-\sigma\omega\,c_1^S-\sigma(1-\omega)\,c_1^H\big] \;=\; \big(C^S\big)^{-\sigma}\big(1-\sigma c_0^S\big).$$

Evaluating at the steady state gives $\beta\tilde{R}\,\Omega=\left(C^S\right)^{-\sigma}$; dividing both sides by this relation and rearranging yields

$$c_0^S \;=\; \mathbb{E}\left[\omega\,c_1^S+(1-\omega)\,c_1^H-\frac{1}{\sigma}\,\tilde{r}_1\right], \tag{B.14}$$

or, solved for the expected future Saver consumption,

$$\mathbb{E}\big[c_1^S\big] \;=\; \frac{1}{\omega}\left(c_0^S+\frac{1}{\sigma}\,\mathbb{E}[\tilde{r}_1]\right)-\frac{1-\omega}{\omega}\,\mathbb{E}\big[c_1^H\big]\,. \tag{B.15}$$

The steady-state return

As established above, the steady-state version of (B.12) is $\beta\tilde{R}\,\Omega=\left(C^S\right)^{-\sigma}$, that is

$$\beta\tilde{R}\left[s+(1-s)\,\zeta^{\sigma}\right] \;=\; 1 \qquad\Longleftrightarrow\qquad s\tilde{R} \;=\; \frac{\omega}{\beta}, \tag{B.16}$$

where the implication follows from multiplying both sides of the first equality by s and using the definition of ω in (B.13). Since $\zeta>1$ implies $\omega<s$, the steady-state return satisfies $\tilde{R}<\frac{1}{\beta}$: the risk of transitioning into the high-marginal-utility HtM state makes households willing to hold bonds at a depressed return. Equation (B.16) is used repeatedly below: the effective gross return on carried wealth, $s\tilde{R}$, equals $\frac{\omega}{\beta}$.

B.4. The Consumption Function of an HtM Household

Log-linearizing the HtM budget constraint (B.9) at $t = 0$ and $t = 1$:

$$c_0^H = \frac{\tilde{Y}^H}{C^H}\,\tilde{y}_0^H + \frac{C^H - \tilde{Y}^H}{C^H}\left(\tilde{r}_0 + \tilde{z}_0^S\right), \tag{B.17}$$

$$\mathbb{E}\left[c_1^H\right] = \frac{\tilde{Y}^H}{C^H}\,\mathbb{E}\left[\tilde{y}_1^H\right] + \frac{C^H - \tilde{Y}^H}{C^H}\,\mathbb{E}\left[\tilde{r}_1 + \tilde{z}_1^S\right]. \tag{B.18}$$

Current HtM consumption is a convex combination of HtM income and the pooled wealth brought in by transitioning Savers, with weights given by the steady-state financing shares. The coefficient $\frac{C^H - \tilde{Y}^H}{C^H}$ is the share of HtM consumption financed by pooled bonds of the transitioning Savers.

B.5. Solving for Expected Future HtM Consumption: the Factor ϕ

Equation (B.18) contains $\mathbb{E}\left[\tilde{z}_1^S\right]$, the bonds carried into period 1 by the Savers. Log-linearizing (B.6):

$$\mathbb{E}\left[\tilde{z}_1^S\right] \;=\; \frac{C^S}{s\tilde{R}\tilde{Z}^S}\,\mathbb{E}\left[c_1^S\right] - \frac{\tilde{Y}^S}{s\tilde{R}\tilde{Z}^S}\,\mathbb{E}\left[\tilde{y}_1^S\right] - \mathbb{E}[\tilde{r}_1]. \tag{B.19}$$

Substituting the Euler equation (B.15) into (B.19):

$$\mathbb{E}\left[\tilde{z}_1^S\right] \quad = \quad \frac{C^S}{s\tilde{R}\tilde{Z}^S}\left[\frac{1}{\omega}\left(c_0^S + \frac{1}{\sigma}\,\mathbb{E}[\tilde{r}_1]\right) - \frac{1-\omega}{\omega}\,\mathbb{E}\left[c_1^H\right]\right] \;-\; \frac{\tilde{Y}^S}{s\tilde{R}\tilde{Z}^S}\,\mathbb{E}\left[\tilde{y}_1^S\right] \;-\; \mathbb{E}[\tilde{r}_1]. \tag{B.20}$$

Substituting (B.20) into (B.18) yields an equation in which $\mathbb{E}\left[c_1^H\right]$ appears on both sides: future HtM consumption depends on the pooled bonds, the pooled bonds depend on the Saver's plan, and the Saver's plan depends on future HtM consumption through the Euler equation. Note that the term $-\mathbb{E}[\tilde{r}_1]$ in (B.20) cancels against the term $+\mathbb{E}[\tilde{r}_1]$ that enters (B.18) directly, leaving

$$\begin{aligned}\mathbb{E}\left[c_1^H\right] \;=\; & \frac{\tilde{Y}^H}{C^H}\,\mathbb{E}\left[\tilde{y}_1^H\right] - \frac{C^H - \tilde{Y}^H}{C^H}\,\frac{\tilde{Y}^S}{s\tilde{R}\tilde{Z}^S}\,\mathbb{E}\left[\tilde{y}_1^S\right] \\ & + \frac{C^H - \tilde{Y}^H}{C^H}\,\frac{C^S}{s\tilde{R}\tilde{Z}^S}\,\frac{1}{\omega}\left(c_0^S + \frac{1}{\sigma}\,\mathbb{E}[\tilde{r}_1]\right) - \frac{C^H - \tilde{Y}^H}{C^H}\,\frac{C^S}{s\tilde{R}\tilde{Z}^S}\,\frac{1-\omega}{\omega}\,\mathbb{E}\left[c_1^H\right].\end{aligned} \tag{B.21}$$

Collecting the $\mathbb{E}\left[c_1^H\right]$ terms on the left-hand side:

$$\begin{aligned}\left(1 + \frac{C^H - \tilde{Y}^H}{C^H}\,\frac{C^S}{s\tilde{R}\tilde{Z}^S}\,\frac{1-\omega}{\omega}\right)\mathbb{E}\left[c_1^H\right] \;=\; & \frac{\tilde{Y}^H}{C^H}\,\mathbb{E}\left[\tilde{y}_1^H\right] \\ & - \frac{C^H - \tilde{Y}^H}{C^H}\,\frac{\tilde{Y}^S}{s\tilde{R}\tilde{Z}^S}\,\mathbb{E}\left[\tilde{y}_1^S\right] + \frac{C^H - \tilde{Y}^H}{C^H}\,\frac{C^S}{s\tilde{R}\tilde{Z}^S}\,\frac{1}{\omega}\left(c_0^S + \frac{1}{\sigma}\,\mathbb{E}[\tilde{r}_1]\right),\end{aligned} \tag{B.22}$$

and dividing through by the bracket solves:

$$\mathbb{E}\left[c_1^H\right] \;=\; \phi\,\frac{\tilde{Y}^H}{C^H}\,\mathbb{E}\left[\tilde{y}_1^H\right] - \phi\,\frac{C^H - \tilde{Y}^H}{C^H}\,\frac{\tilde{Y}^S}{s\tilde{R}\tilde{Z}^S}\,\mathbb{E}\left[\tilde{y}_1^S\right] + \mathcal{E}\left(c_0^S + \frac{1}{\sigma}\,\mathbb{E}[\tilde{r}_1]\right), \tag{B.23}$$

where

$$\phi \;\equiv\; \frac{1}{1+\dfrac{C^H-\tilde{Y}^H}{C^H}\,\dfrac{C^S}{s\tilde{R}\tilde{Z}^S}\,\dfrac{1-\omega}{\omega}} \;\in\; (0,1) \;=\; \frac{\omega\,\dfrac{s\tilde{R}\tilde{Z}^S}{C^S}}{\omega\,\dfrac{s\tilde{R}\tilde{Z}^S}{C^S}+(1-\omega)\,\dfrac{C^H-\tilde{Y}^H}{C^H}} \tag{B.24}$$

and

$$\mathcal{E} \;\equiv\; \phi\,\frac{C^H-\tilde{Y}^H}{C^H}\,\frac{C^S}{s\tilde{R}\tilde{Z}^S}\,\frac{1}{\omega}. \tag{B.25}$$

The factor ϕ resolves the fixed point: it is the attenuation applied to every driver of future HtM consumption once the feedback through the Savers' plan is accounted for. Its first expression in (B.24) shows its structure: the two terms in the denominator are the Euler-weighted consumption-financing services of Saver bonds in the future Saver and HtM states, and ϕ is the share of those services accruing in the Saver's own continuation state.

Two algebraic identities involving ϕ and $\mathcal{E}$ carry the simplification of the consumption function.

Lemma B.1. *The main identity:* $1-(1-\omega)\,\mathcal{E} \;=\; \phi$.

Proof. Let $x \equiv \frac{C^H-\tilde{Y}^H}{C^H}\,\frac{C^S}{s\tilde{R}\tilde{Z}^S}$, so that $\mathcal{E} = \phi\,x/\omega$ and $\phi = \omega/\big(\omega+x(1-\omega)\big)$. Then

$$1-(1-\omega)\,\mathcal{E} \;=\; 1-\frac{(1-\omega)\,x}{\omega+x(1-\omega)} \;=\; \frac{\omega}{\omega+x(1-\omega)} \;=\; \phi.$$

□

Lemma B.2. *Corollary to the main identity:* $\dfrac{1}{\sigma\omega}\left[(1-\omega)\,\mathcal{E}-(1-\sigma\omega)\right] \;=\; 1-\dfrac{\phi}{\sigma\omega}$.

Proof. By Lemma B.1, $(1-\omega)\,\mathcal{E} = 1-\phi$, so the bracket equals $(1-\phi)-(1-\sigma\omega) = \sigma\omega-\phi$; dividing by $\sigma\omega$ gives the result. □

B.6. The Consumption Function of a Saver

The consumption function is obtained from the log-linearized constraint (B.11) in three substitution steps.

Step 1: eliminating c_0^H. Substituting c_0^H (B.17) into the aggregate constraint for bonds (B.11):

$$c_0^S+\frac{1}{s\tilde{R}}\,\mathbb{E}\big[c_1^S-\tilde{r}_1\big] \;=\; \left(\frac{\tilde{R}\tilde{Z}^S}{C^S}-\frac{\lambda}{1-\lambda}\,\frac{C^H-\tilde{Y}^H}{C^S}\right)\big(\tilde{r}_0+\tilde{z}_0^S\big) + \frac{\tilde{Y}^S}{C^S}\,\tilde{y}_0^S+\frac{\tilde{Y}^S}{s\tilde{R}\,C^S}\,\mathbb{E}\big[\tilde{y}_1^S-\tilde{r}_1\big]. \tag{B.26}$$

A share $\frac{\lambda}{1-\lambda}\frac{C^H-\tilde{Y}^H}{C^S}$ of the funds received at $t=0$ is redistributed to the HtM group.

Step 2: eliminating $\mathbb{E}\left[c_1^S\right]$. Substituting the Euler equation (B.15) into (B.26) gives

$$c_0^S + \frac{1}{s\tilde{R}}\left[\frac{1}{\omega}\left(c_0^S + \frac{1}{\sigma}\,\mathbb{E}[\tilde{r}_1]\right) - \frac{1-\omega}{\omega}\,\mathbb{E}\left[c_1^H\right] - \mathbb{E}[\tilde{r}_1]\right] = \left(\frac{\tilde{R}\tilde{Z}^S}{C^S} - \frac{\lambda}{1-\lambda}\,\frac{C^H - \tilde{Y}^H}{C^S}\right)\left(\tilde{r}_0 + \tilde{z}_0^S\right) + \frac{\tilde{Y}^S}{C^S}\,\tilde{y}_0^S + \frac{\tilde{Y}^S}{s\tilde{R}\,C^S}\,\mathbb{E}\left[\tilde{y}_1^S - \tilde{r}_1\right]. \quad \text{(B.27)}$$

Collecting terms in c_0^S then yields

$$\left(1 + \frac{1}{s\tilde{R}\omega}\right)c_0^S = \frac{1}{s\tilde{R}}\,\frac{1-\omega}{\omega}\,\mathbb{E}\left[c_1^H\right] - \frac{1}{s\tilde{R}}\,\frac{1-\sigma\omega}{\sigma\omega}\,\mathbb{E}[\tilde{r}_1] + \left(\frac{\tilde{R}\tilde{Z}^S}{C^S} - \frac{\lambda}{1-\lambda}\,\frac{C^H - \tilde{Y}^H}{C^S}\right)\left(\tilde{r}_0 + \tilde{z}_0^S\right) + \frac{\tilde{Y}^S}{C^S}\,\tilde{y}_0^S + \frac{\tilde{Y}^S}{s\tilde{R}\,C^S}\,\mathbb{E}\left[\tilde{y}_1^S - \tilde{r}_1\right], \quad \text{(B.28)}$$

where the coefficient on $\mathbb{E}[\tilde{r}_1]$ in the first line collects the term $-\frac{1}{s\tilde{R}}\frac{1}{\sigma\omega}$ brought in by the Euler equation with the term $+\frac{1}{s\tilde{R}}$ from the left-hand side of (B.26).

Step 3: eliminating $\mathbb{E}\left[c_1^H\right]$. Substituting (B.23) into (B.28) and collecting the terms in c_0^S on the left-hand side yields the consumption function in its unsimplified form:

$$\begin{aligned} c_0^S\left[1 + \frac{1}{s\tilde{R}\omega} - \frac{1-\omega}{s\tilde{R}\omega}\,\mathcal{E}\right] = {} & \left(\frac{\tilde{R}\tilde{Z}^S}{C^S} - \frac{\lambda}{1-\lambda}\,\frac{C^H - \tilde{Y}^H}{C^S}\right)\left(\tilde{r}_0 + \tilde{z}_0^S\right) \\ & - \frac{1}{s\tilde{R}}\,\frac{1-\sigma\omega}{\sigma\omega}\,\mathbb{E}[\tilde{r}_1] \\ & + \frac{1-\omega}{s\tilde{R}\omega}\,\phi\,\frac{\tilde{Y}^H}{C^H}\,\mathbb{E}\left[\tilde{y}_1^H\right] \\ & - \frac{1-\omega}{s\tilde{R}\omega}\,\phi\,\frac{C^H - \tilde{Y}^H}{C^H}\,\frac{\tilde{Y}^S}{s\tilde{R}\tilde{Z}^S}\,\mathbb{E}\left[\tilde{y}_1^S\right] \\ & + \frac{1}{\sigma}\,\frac{1-\omega}{s\tilde{R}\omega}\,\mathcal{E}\,\mathbb{E}[\tilde{r}_1] \\ & + \frac{\tilde{Y}^S}{C^S}\,\tilde{y}_0^S + \frac{\tilde{Y}^S}{s\tilde{R}\,C^S}\,\mathbb{E}\left[\tilde{y}_1^S - \tilde{r}_1\right], \end{aligned} \quad \text{(B.29)}$$

where the left-hand-side bracket and line 5 use the definition of $\mathcal{E}$ in (B.25).

Step 4: simplifying. In (B.29), the terms in $\mathbb{E}[\tilde{r}_1]$ and $\mathbb{E}\left[\tilde{y}_1^S\right]$ are dispersed across several lines, and $\mathcal{E}$ appears both in the left-hand-side bracket and on the right. The coefficients combine through Lemmas B.1 and B.2, component by component.

The left-hand-side bracket

Applying Lemma B.1 identity directly:

$$1 + \frac{1}{s\tilde{R}\omega} - \frac{1-\omega}{s\tilde{R}\omega}\,\mathcal{E} = 1 + \frac{1}{s\tilde{R}\omega}\underbrace{\left[1 - (1-\omega)\,\mathcal{E}\right]}_{=\phi} = 1 + \frac{\phi}{s\tilde{R}\omega}. \quad \text{(B.30)}$$

The $\mathbb{E}[\tilde{r}_1]$ terms

Three lines of (B.29) contribute to the coefficient on $\mathbb{E}[\tilde{r}_1]$: the direct Euler term $-\frac{1}{s\tilde{R}}\frac{1-\sigma\omega}{\sigma\omega}$, the feedback term $+\frac{1}{\sigma}\frac{1-\omega}{s\tilde{R}\omega}\mathcal{E}$ from future HtM consumption, and the term $-\frac{\tilde{Y}^S}{s\tilde{R}C^S}$ from the income line. Using Lemma B.2, the first two sum to

$$\frac{1}{s\tilde{R}}\cdot\frac{1}{\sigma\omega}\Big[(1-\omega)\,\mathcal{E}-(1-\sigma\omega)\Big] \;=\; \frac{1}{s\tilde{R}}\left(1-\frac{\phi}{\sigma\omega}\right) \;=\; -\frac{1}{s\tilde{R}}\cdot\frac{\phi-\sigma\omega}{\sigma\omega}, \tag{B.31}$$

so that, adding the discounting term, the combined coefficient on $\mathbb{E}[\tilde{r}_1]$ is

$$-\frac{1}{s\tilde{R}}\left(\frac{\tilde{Y}^S}{C^S}+\frac{\phi-\sigma\omega}{\sigma\omega}\right).$$

The $\mathbb{E}\big[\tilde{y}_1^S\big]$ terms

Two lines contribute. Using the definition of $\mathcal{E}$ in (B.25), the term originating from HtM consumption becomes

$$-\frac{1-\omega}{s\tilde{R}\omega}\,\phi\,\frac{C^H-\tilde{Y}^H}{C^H}\,\frac{\tilde{Y}^S}{s\tilde{R}\tilde{Z}^S} \;=\; -\frac{\tilde{Y}^S}{s\tilde{R}C^S}\,(1-\omega)\,\mathcal{E},$$

and adding the direct present-value term $+\frac{\tilde{Y}^S}{s\tilde{R}C^S}$ from the income line, Lemma B.1 yields

$$\frac{\tilde{Y}^S}{s\tilde{R}C^S}\big[1-(1-\omega)\,\mathcal{E}\big] \;=\; \frac{\tilde{Y}^S}{C^S}\cdot\frac{\phi}{s\tilde{R}}. \tag{B.32}$$

The $\mathbb{E}\big[\tilde{y}_1^H\big]$ term

The coefficient $\frac{1-\omega}{s\tilde{R}\omega}\,\phi\,\frac{\tilde{Y}^H}{C^H}$ is unaffected by the simplification.

The simplified equation

Applying these combinations to (B.29):

$$\begin{aligned} c_0^S\left(1+\frac{\phi}{s\tilde{R}\omega}\right) \;=\;& \frac{\tilde{Y}^S}{C^S}\left(\tilde{y}_0^S+\frac{\phi}{s\tilde{R}}\,\mathbb{E}\big[\tilde{y}_1^S\big]\right) \\ &-\frac{1}{s\tilde{R}}\left(\frac{\tilde{Y}^S}{C^S}+\frac{\phi-\sigma\omega}{\sigma\omega}\right)\mathbb{E}[\tilde{r}_1] \\ &+\frac{1}{C^S}\left[\tilde{R}\tilde{Z}^S-\frac{\lambda}{1-\lambda}\big(C^H-\tilde{Y}^H\big)\right]\big(\tilde{r}_0+\tilde{z}_0^S\big) \\ &+\frac{1-\omega}{s\tilde{R}\omega}\,\phi\,\frac{\tilde{Y}^H}{C^H}\,\mathbb{E}\big[\tilde{y}_1^H\big]\,. \end{aligned} \tag{B.33}$$

Dividing through by the left-hand-side bracket (B.30), the consumption function of a Saver at $t = 0$ is

$$\begin{aligned} c_0^S \;=\; & \left(\frac{s\tilde{R}\omega}{\phi + s\tilde{R}\omega}\right)\frac{\tilde{Y}^S}{C^S}\left(\tilde{y}_0^S + \frac{\phi}{s\tilde{R}}\,\mathbb{E}\big[\tilde{y}_1^S\big]\right) \\ & - \left(\frac{s\tilde{R}\omega}{\phi + s\tilde{R}\omega}\right)\frac{1}{s\tilde{R}}\left(\frac{\tilde{Y}^S}{C^S} + \frac{\phi - \sigma\omega}{\sigma\omega}\right)\mathbb{E}[\tilde{r}_1] \\ & + \left(\frac{s\tilde{R}\omega}{\phi + s\tilde{R}\omega}\right)\frac{1}{C^S}\left[\tilde{R}\tilde{Z}^S - \frac{\lambda}{1-\lambda}\big(C^H - \tilde{Y}^H\big)\right]\big(\tilde{r}_0 + \tilde{z}_0^S\big) \\ & + \left(\frac{s\tilde{R}\omega}{\phi + s\tilde{R}\omega}\right)\frac{\tilde{Y}^H}{C^H}\,\frac{1-\omega}{\omega}\,\frac{\phi}{s\tilde{R}}\,\mathbb{E}\big[\tilde{y}_1^H\big]\,. \end{aligned} \tag{B.34}$$

The prefactor in terms of $\mathcal{F}$. Define

$$\mathcal{F} \;\equiv\; \omega + (1-\omega)\,\frac{C^H - \tilde{Y}^H}{C^H}\,\frac{C^S}{s\tilde{R}\tilde{Z}^S} \;=\; \frac{\omega}{\phi}. \tag{B.35}$$

Using the steady-state return (B.16), $s\tilde{R} = \frac{\omega}{\beta}$, the prefactor of (B.34) becomes

$$\frac{s\tilde{R}\omega}{\phi + s\tilde{R}\omega} \;=\; \frac{\frac{\omega^2}{\beta}}{\phi + \frac{\omega^2}{\beta}} \;=\; \frac{\omega^2}{\beta\phi + \omega^2} \;=\; \frac{\frac{\omega^2}{\phi}}{\beta + \frac{\omega^2}{\phi}} \;=\; \frac{\omega\mathcal{F}}{\omega\mathcal{F} + \beta}, \tag{B.36}$$

which is the form used in the main text. The consumption function (B.34) with the prefactor (B.36) is equation (23) of Section 3.

B.7. The Benchmark with Absorbing States

The benchmark sets $s = h = 1$: no household ever changes state. Then $C^H = \tilde{Y}^H$ (the HtM consume their income), $\omega = \phi = \mathcal{F} = 1$, and the steady-state Euler equation (B.16) gives $\tilde{R} = \frac{1}{\beta}$. The Saver's problem is a standard two-period consumption-saving problem. Its intertemporal budget constraint, log-linearized, is (B.11) with $s = 1$ and no HtM transfer term; the Euler equation is $c_0^S = \mathbb{E}\big[c_1^S - \frac{1}{\sigma}\tilde{r}_1\big]$. Substituting the Euler equation into the constraint and collecting terms — the coefficient on $\mathbb{E}[\tilde{r}_1]$ collects $-\frac{1}{\tilde{R}}\frac{\tilde{Y}^S}{C^S}$ from the discounting of future income and $-\frac{1}{\tilde{R}}\frac{1-\sigma}{\sigma}$ from the Euler substitution — the consumption function is

$$\begin{aligned} c_0^S \;=\; & \left(\frac{1}{1+\beta}\right)\frac{\tilde{Y}^S}{C^S}\left(\tilde{y}_0^S + \frac{1}{\tilde{R}}\,\mathbb{E}\big[\tilde{y}_1^S\big]\right) \\ & - \left(\frac{1}{1+\beta}\right)\frac{1}{\tilde{R}}\left(\frac{\tilde{Y}^S}{C^S} + \frac{1-\sigma}{\sigma}\right)\mathbb{E}[\tilde{r}_1] \\ & + \left(\frac{1}{1+\beta}\right)\frac{1}{C^S}\,\tilde{R}\tilde{Z}^S\left(\tilde{r}_0 + \tilde{z}_0^S\right), \end{aligned} \tag{B.37}$$

which is equation (24) of Section 3. It coincides with the absorbing-states limit of (B.34), where $s \to 1$, $\omega, \phi, \mathcal{F} \to 1$, and $C^H = \tilde{Y}^H$: in particular $\frac{\phi - \sigma\omega}{\sigma\omega} \to \frac{1-\sigma}{\sigma}$, so the two interest-rate coefficients agree at the limit.

B.8. Auxiliary Results Used in Section 3

This section collects the results invoked in the interpretation of the consumption functions.

B.8.1. *The wealth base*

From the HtM budget constraint (B.9), $\frac{\lambda}{1-\lambda}\left(C^H - \tilde{Y}^H\right) = (1-s)\,\tilde{R}\tilde{Z}^S$, so the wealth base in the third line of (B.34) simplifies to

$$\tilde{R}\tilde{Z}^S - \frac{\lambda}{1-\lambda}\left(C^H - \tilde{Y}^H\right) \;=\; \tilde{R}\tilde{Z}^S - (1-s)\,\tilde{R}\tilde{Z}^S \;=\; s\,\tilde{R}\tilde{Z}^S : \tag{B.38}$$

transition risk taxes the Savers' gross wealth income at the rate $1-s$, regardless of how much HtM consumption the transfer finances.

B.8.2. *Level MPCs*

The consumption functions are expressed in log-deviations, so their coefficients are elasticities. The corresponding level responses are obtained by multiplying by the steady-state level of consumption and dividing by the steady-state level of the shocked variable. For a shock to current Saver income:

$$\frac{\partial C_0^S}{\partial \tilde{Y}_0^S} \;=\; \frac{C^S}{\tilde{Y}^S}\cdot\frac{\partial c_0^S}{\partial \tilde{y}_0^S} \;=\; \frac{C^S}{\tilde{Y}^S}\cdot\frac{\omega\mathcal{F}}{\omega\mathcal{F}+\beta}\,\frac{\tilde{Y}^S}{C^S} \;=\; \frac{\omega\mathcal{F}}{\omega\mathcal{F}+\beta}, \tag{B.39}$$

and analogously $\frac{1}{1+\beta}$ in the benchmark: the level MPC out of current income is the prefactor itself, with the income share cancelling out. The comparison of prefactors is therefore a comparison of marginal propensities in levels and requires no assumption on the relative steady states.

B.8.3. *The discount on expected future income*

Using $s\tilde{R} = \frac{\omega}{\beta}$ from (B.16) and $\phi = \frac{\omega}{\mathcal{F}}$ from (B.35):

$$\frac{\phi}{s\tilde{R}} \;=\; \frac{\beta\phi}{\omega} \;=\; \frac{\beta}{\mathcal{F}}, \tag{B.40}$$

against $\frac{1}{\tilde{R}} = \beta$ in the benchmark. Combining with the prefactor, the level iMPC out of expected future Saver income is

$$\frac{\partial C_0^S}{\partial\,\mathbb{E}\left[\tilde{Y}_1^S\right]} \;=\; \frac{\omega\mathcal{F}}{\omega\mathcal{F}+\beta}\cdot\frac{\beta}{\mathcal{F}} \;=\; \frac{\omega\beta}{\omega\mathcal{F}+\beta}, \tag{B.41}$$

and the comparison with the benchmark value $\frac{\beta}{1+\beta}$ reduces to

$$\frac{\omega\beta}{\omega\mathcal{F}+\beta} \;\gtrless\; \frac{\beta}{1+\beta} \quad\Longleftrightarrow\quad \omega\,(1+\beta) \;\gtrless\; \omega\mathcal{F}+\beta \quad\Longleftrightarrow\quad \omega\,(1-\mathcal{F}) \;\gtrless\; \beta\,(1-\omega). \tag{B.42}$$

B.8.4. *The interest-rate coefficient*

Using again $\phi = \frac{\omega}{\mathcal{F}}$, the CRRA income–substitution component of the interest-rate coefficient in (B.34) can be written

$$\frac{\phi - \sigma\omega}{\sigma\omega} \;=\; \frac{\phi}{\sigma\omega} - 1 \;=\; \frac{1}{\sigma\mathcal{F}} - 1, \tag{B.43}$$

which is the benchmark component $\frac{1-\sigma}{\sigma} = \frac{1}{\sigma} - 1$ with the substitution elasticity $\frac{1}{\sigma}$ replaced by $\frac{1}{\sigma\mathcal{F}}$: the factor $\frac{1}{\mathcal{F}} = \frac{\phi}{\omega}$ applies to the intertemporal-substitution margin in the same way that it applies to expected future income in (B.40).

B.8.5. *The relative loading on expected HtM income*

Converting the coefficients on $\mathbb{E}\big[\tilde{y}_1^H\big]$ and $\mathbb{E}\big[\tilde{y}_1^S\big]$ in (B.34) to level responses (each coefficient multiplied by C^S and divided by the level of the respective income):

$$\frac{\partial C_0^S}{\partial\,\mathbb{E}\big[\tilde{Y}_1^H\big]} = C^S \cdot \frac{\omega\mathcal{F}}{\omega\mathcal{F}+\beta}\,\frac{\tilde{Y}^H}{C^H}\,\frac{1-\omega}{\omega}\,\frac{\phi}{s\tilde{R}} \cdot \frac{1}{\tilde{Y}^H} = \frac{\omega\mathcal{F}}{\omega\mathcal{F}+\beta}\,\frac{\phi}{s\tilde{R}} \cdot \frac{C^S}{C^H}\,\frac{1-\omega}{\omega}, \tag{B.44}$$

while $\frac{\partial C_0^S}{\partial\,\mathbb{E}\big[\tilde{Y}_1^S\big]}$ is given in (B.41). Taking the ratio, the prefactor and the discount cancel:

$$\frac{\partial C_0^S/\partial\,\mathbb{E}\big[\tilde{Y}_1^H\big]}{\partial C_0^S/\partial\,\mathbb{E}\big[\tilde{Y}_1^S\big]} \;=\; \zeta\,\frac{1-\omega}{\omega} \;=\; \frac{(1-s)\,\zeta^{\sigma+1}}{s}, \tag{B.45}$$

where the last equality uses $\frac{1-\omega}{\omega} = \frac{(1-s)\,\zeta^{\sigma}}{s}$ from (B.13). The exponent decomposes as follows: ζ^{σ} enters through the Euler-equation weight (the marginal utility of the constrained state), and the remaining factor ζ through the conversion to levels.

B.8.6. *The thresholds on $\mathcal{F}$ in terms of primitives*

Both thresholds run through the ratio $\frac{C^H-\tilde{Y}^H}{C^H}\,\frac{C^S}{s\tilde{R}\tilde{Z}^S}$ appearing in the definition of $\mathcal{F}$, (B.35). This ratio has a simple expression in primitives.

Lemma B.3. *The absorption ratio in primitives:* $\dfrac{C^H - \tilde{Y}^H}{C^H}\,\dfrac{C^S}{s\tilde{R}\tilde{Z}^S} \;=\; \dfrac{(1-h)\,\zeta}{s}.$

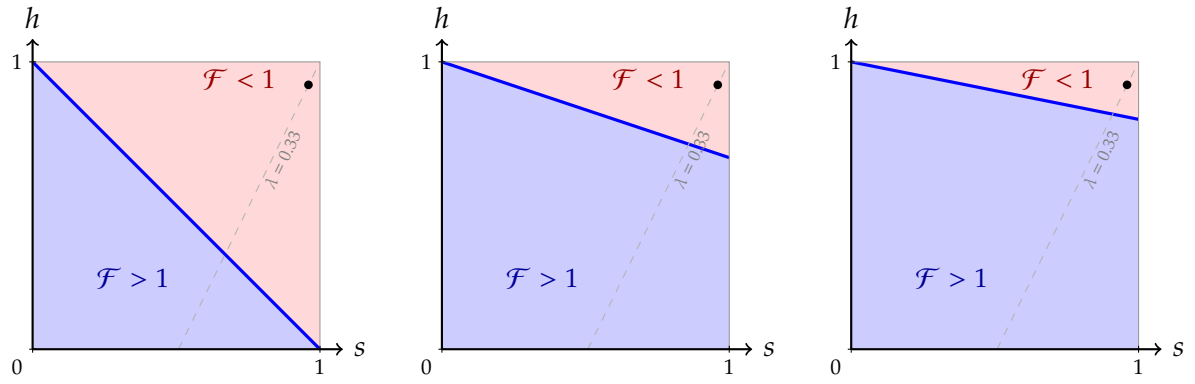
Proof. The HtM budget constraint (B.9) and the stationarity condition (B.1) give

$$C^H - \tilde{Y}^H = \frac{(1-\lambda)(1-s)}{\lambda}\,\tilde{R}\tilde{Z}^S = (1-h)\,\tilde{R}\tilde{Z}^S.$$

Substituting,

$$\frac{C^H - \tilde{Y}^H}{C^H}\,\frac{C^S}{s\tilde{R}\tilde{Z}^S} \;=\; \frac{(1-h)\,\tilde{R}\tilde{Z}^S}{C^H} \cdot \frac{C^S}{s\tilde{R}\tilde{Z}^S} \;=\; \frac{(1-h)}{s} \cdot \frac{C^S}{C^H} \;=\; \frac{(1-h)\,\zeta}{s},$$

with $\tilde{R}$ and $\tilde{Z}^S$ cancelling: the ratio is independent of the level of liquid assets. □



(a) $\zeta = 1$ **(b)** $\zeta = 3$ **(c)** $\zeta = 5$

Figure 2: Parameter regions in the (s, h)-plane for the tilt of the Saver's consumption response, for different levels of steady-state inequality $\zeta = C^S/C^H$.

Notes: The solid line is the threshold $\mathcal{F} = 1$. In the blue region front-loading is active ($\mathcal{F} > 1$) and the response profile tilts towards current income; in the red region $\mathcal{F} < 1$. The dashed line is the set of (s, h) consistent with a HtM share $\lambda = 0.33$ under the stationarity condition; the marker is the quarterly calibration of Bilbiie (2025), $s = 0.96$ with implied $h = 0.92$, which lies in the $\mathcal{F} < 1$ region for every ζ considered.

The condition $\mathcal{F} > 1$. The definition (B.35) writes $\mathcal{F}$ as a convex combination of 1 and the absorption ratio $\frac{C^H - \tilde{Y}^H}{C^H} \frac{C^S}{s\tilde{R}\tilde{Z}^S}$,

$$\mathcal{F} \;=\; \omega \cdot 1 + (1-\omega) \cdot \frac{C^H - \tilde{Y}^H}{C^H} \frac{C^S}{s\tilde{R}\tilde{Z}^S}, \qquad \omega \in (0,1),$$

so $\mathcal{F}$ lies strictly between 1 and the ratio, and exceeds one if and only if the ratio does. By Lemma B.3:

$$\mathcal{F} > 1 \quad \Longleftrightarrow \quad \frac{C^H - \tilde{Y}^H}{C^H} \frac{C^S}{s\tilde{R}\tilde{Z}^S} > 1 \quad \Longleftrightarrow \quad (1-h)\,\zeta > s. \tag{B.46}$$

In the terminology of Section 3, (B.46) is the condition under which front-loading is *active* and the consumption profile tilts towards the present. It involves neither σ, β, nor the level of liquid assets. In the (s, h)-plane its boundary is the line $h^*(s) = 1 - \frac{s}{\zeta}$, shown in Figure 2.

The condition $\omega\mathcal{F} > 1$. Substituting the definition (B.35) and writing the absorption ratio in its primitive form $\frac{(1-h)\zeta}{s}$ of Lemma B.3:

$$\begin{aligned}
\omega\mathcal{F} > 1 \quad &\Longleftrightarrow \quad \omega^2 + \omega(1-\omega)\frac{(1-h)\,\zeta}{s} > 1 \\
&\Longleftrightarrow \quad \omega(1-\omega)\frac{(1-h)\,\zeta}{s} > (1-\omega)(1+\omega) \\
&\Longleftrightarrow \quad \frac{(1-h)\,\zeta}{s} > \frac{1+\omega}{\omega} \\
&\Longleftrightarrow \quad \frac{(1-h)\,\zeta}{s} > 1 + \frac{1}{\omega},
\end{aligned}$$

where dividing by $\omega(1-\omega) > 0$ uses $\omega \in (0,1)$. Since $\frac{1}{\omega} > 1$, the hurdle $1 + \frac{1}{\omega}$ exceeds 2, and the condition is strictly stronger than (B.46). Using $\frac{1}{\omega} = \frac{s+(1-s)\zeta^\sigma}{s}$:

$$\begin{aligned} \omega\mathcal{F} > 1 \quad &\Longleftrightarrow \quad \frac{(1-h)\,\zeta}{s} > \frac{2s + (1-s)\,\zeta^\sigma}{s} \\ &\Longleftrightarrow \quad (1-h)\,\zeta > 2s + (1-s)\,\zeta^\sigma. \end{aligned}$$

In the terminology of Section 3, this is the condition under which front-loading *prevails* over self-insurance, and the Saver's level MPC out of current income exceeds the absorbing-states benchmark.

Appendix C. Derivation of the Saver's Consumption Function in Infinite-Horizon THANK

This appendix derives the infinite-horizon Saver consumption function in a THANK economy with positive steady-state liquidity, $\tilde{Z}^S > 0$, and unequal steady-state consumption, $C^S > C^H$, by the characteristic-root method. Since liquidity is positive, all variables — including per-Saver bonds — are log deviations from the steady state:

$$C_t^j = C^j e^{c_t^j}, \qquad \tilde{Y}_t^j = \tilde{Y}^j e^{\tilde{y}_t^j}, \qquad \tilde{R}_t = \tilde{R} e^{\tilde{r}_t}, \qquad \tilde{Z}_t^S = \tilde{Z}^S e^{\tilde{z}_t^S}, \qquad j \in \{S, H\}. \tag{C.1}$$

Aggregate liquid assets are $B_t = (1-\lambda)\tilde{Z}_t^S$, so $\tilde{z}_t^S$ is also the log deviation of aggregate liquidity.

The marginal-utility weight and the steady-state Euler equation are, just like in the two-period model:

$$\omega = \frac{s}{s + (1-s)\zeta^\sigma}, \qquad \beta\tilde{R} = \frac{1}{s + (1-s)\zeta^\sigma}, \qquad s\tilde{R} = \frac{\omega}{\beta}, \qquad \zeta \equiv \frac{C^S}{C^H} > 1. \tag{C.2}$$

C.1. Budget Constraints

As in the two-period model, bonds are pooled within groups: a continuing Saver retains the fraction s of per-Saver bonds, and the HtM group receives, per member, $\frac{(1-\lambda)(1-s)}{\lambda}\tilde{Z}_t^S = (1-h)\tilde{Z}_t^S$ by the stationarity of the population shares, $(1-\lambda)(1-s) = \lambda(1-h)$. The level budget constraints are therefore

$$C_t^H = \tilde{Y}_t^H + (1-h)\,\tilde{R}_t\,\tilde{Z}_t^S, \tag{C.3}$$

$$C_t^S + \tilde{Z}_{t+1}^S = \tilde{Y}_t^S + s\,\tilde{R}_t\,\tilde{Z}_t^S. \tag{C.4}$$

Evaluated at the steady state:

$$C^H - \tilde{Y}^H = (1-h)\,\tilde{R}\tilde{Z}^S, \qquad C^S - \tilde{Y}^S = \left(s\tilde{R} - 1\right)\tilde{Z}^S. \tag{C.5}$$

Log-linearizing (C.3) and (C.4) around the steady state:

$$c_t^H = \frac{\tilde{Y}^H}{C^H}\,\tilde{y}_t^H + \alpha_H\left(\tilde{r}_t + \tilde{z}_t^S\right), \qquad \alpha_H \equiv \frac{C^H - \tilde{Y}^H}{C^H}, \tag{C.6}$$

$$c_t^S = \frac{\tilde{Y}^S}{C^S}\,\tilde{y}_t^S + \alpha_S\left(\tilde{r}_t + \tilde{z}_t^S\right) - \alpha_B\,\tilde{z}_{t+1}^S, \qquad \alpha_S \equiv \frac{s\tilde{R}\tilde{Z}^S}{C^S}, \quad \alpha_B \equiv \frac{\tilde{Z}^S}{C^S}, \tag{C.7}$$

with $\alpha_S = s\tilde{R}\,\alpha_B$. In contrast with the zero-liquidity case, the interest rate now enters the budget constraints at first order: $\tilde{r}_t$ revalues the outstanding stock. The coefficients α_H and α_S are the two consumption-financing shares of the two-period analysis.

C.2. Euler Equation and Liquid-Asset Dynamics

The log-linearized CRRA Euler equation of a Saver is

$$c_t^S = \mathbb{E}_t\left[\omega c_{t+1}^S + (1-\omega)c_{t+1}^H - \frac{1}{\sigma}\tilde{r}_{t+1}\right]. \tag{C.8}$$

Substituting (C.7) and (C.6) into (C.8), and collecting the asset terms on the left-hand side:

$$\omega\alpha_B\,\mathbb{E}_t\left[\tilde{z}_{t+2}^S\right] - \left[\alpha_B + \omega\alpha_S + (1-\omega)\alpha_H\right]\tilde{z}_{t+1}^S + \alpha_S\,\tilde{z}_t^S = F_t^0, \tag{C.9}$$

where the forcing term now contains current and expected interest rates:

$$F_t^0 \equiv \omega\,\frac{\tilde{Y}^S}{C^S}\,\mathbb{E}_t\left[\tilde{y}_{t+1}^S\right] + (1-\omega)\,\frac{\tilde{Y}^H}{C^H}\,\mathbb{E}_t\left[\tilde{y}_{t+1}^H\right] - \frac{\tilde{Y}^S}{C^S}\,\tilde{y}_t^S + \left[\omega\alpha_S + (1-\omega)\alpha_H - \frac{1}{\sigma}\right]\mathbb{E}_t[\tilde{r}_{t+1}] - \alpha_S\,\tilde{r}_t. \tag{C.10}$$

Dividing (C.9) by $\omega\alpha_B$:

$$\mathbb{E}_t\left[\tilde{z}_{t+2}^S\right] - \mathcal{A}_\zeta\,\tilde{z}_{t+1}^S + \beta^{-1}\tilde{z}_t^S = F_t, \qquad F_t \equiv \frac{F_t^0}{\omega\alpha_B}, \tag{C.11}$$

since $\frac{\alpha_S}{\omega\alpha_B} = \frac{s\tilde{R}}{\omega} = \beta^{-1}$ by (C.2), and

$$\mathcal{A}_\zeta \equiv \frac{\alpha_B + \omega\alpha_S + (1-\omega)\alpha_H}{\omega\alpha_B} = \frac{1}{\omega} + \frac{\omega}{\beta} + \frac{(1-\omega)}{\beta}\,\frac{(1-h)\,\zeta}{s}, \tag{C.12}$$

where the second equality uses $\alpha_S/\alpha_B = s\tilde{R} = \omega/\beta$ and $\alpha_H/\alpha_B = \frac{C^H - \tilde{Y}^H}{C^H}\,\frac{C^S}{\tilde{Z}^S} = (1-h)\tilde{R}\,\zeta$ from (C.5).

The level of liquidity has cancelled from the homogeneous dynamics, and appears only in the forcing term (C.10), through the interest-rate revaluation.

C.3. Characteristic Equation

The characteristic equation is obtained from the homogeneous part of (C.11). Then, asset dynamics satisfy

$$\mathbb{E}_t\left[\tilde{z}_{t+2}^S\right] - \mathcal{A}_\zeta\tilde{z}_{t+1}^S + \beta^{-1}\tilde{z}_t^S = 0. \tag{C.13}$$

Guessing a solution of the form $\tilde{z}_t^S = K\mu^t$ gives

$$K\mu^{t+2} - \mathcal{A}_\zeta K\mu^{t+1} + \beta^{-1}K\mu^t = 0. \tag{C.14}$$

Dividing by $K\mu^t$ gives the characteristic equation

$$\mu^2 - \mathcal{A}_\zeta \mu + \beta^{-1} = 0. \tag{C.15}$$

Let μ_1 and μ_2 denote the two roots of (C.15), ordered so that $\mu_1 < \mu_2$. By the quadratic formula,

$$\mu_1 = \frac{\mathcal{A}_\zeta - \sqrt{\mathcal{A}_\zeta^2 - 4\beta^{-1}}}{2}, \qquad \mu_2 = \frac{\mathcal{A}_\zeta + \sqrt{\mathcal{A}_\zeta^2 - 4\beta^{-1}}}{2} \tag{C.16}$$

and the quadratic polynomial can be written equivalently as

$$\mu^2 - \mathcal{A}_\zeta \mu + \beta^{-1} = (\mu - \mu_1)(\mu - \mu_2) = \mu^2 - (\mu_1 + \mu_2)\mu + \mu_1\mu_2 \tag{C.17}$$

therefore matching coefficients gives

$$\mu_1 + \mu_2 = \mathcal{A}_\zeta, \qquad \mu_1\mu_2 = \beta^{-1}. \tag{C.18}$$

The first identity says that $\mathcal{A}_\zeta$ governs the sum of the two roots. The second identity says that intertemporal discounting governs their product through β^{-1}.

The saddle configuration $\mu_1 < 1 < \mu_2$ is satisfied if and only if

$$\frac{(1-h)\zeta}{s} > 1 - \frac{\beta}{\omega}. \tag{C.19}$$

In standard calibrations this condition is typically satisfied because $\omega < \beta$. In that case, the right-hand side of (C.19) is negative, while the left-hand side is clearly always positive: $h, s \in (0,1)$ and $\zeta > 0$.

To see why (C.19) holds, one has from the characteristic equation (C.16) that $p(\mu) \equiv \mu^2 - \mathcal{A}_\zeta \mu + \beta^{-1}$ and then $p(0) = \beta^{-1} > 0$. Hence, a root lies between zero and one if and only if $p(1) < 0$, then:

$$\begin{aligned} p(1) < 0 \quad &\iff \quad \mathcal{A}_\zeta > 1 + \beta^{-1} \\ &\iff \quad \frac{1}{\omega} + \frac{\omega}{\beta} + \frac{1-\omega}{\beta}\frac{(1-h)\zeta}{s} > 1 + \frac{1}{\beta} \\ &\iff \quad \frac{(1-h)\zeta}{s} > 1 - \frac{\beta}{\omega}. \end{aligned}$$

Since $\mu_1\mu_2 = \beta^{-1} > 1$, if $\mu_1 < 1$ then the other root must satisfy $\mu_2 > 1$.

C.4. Forward Solution

Using the two identities (C.18) that hold for the roots, the equation for asset dynamics (C.11) can be factored as

$$\mathbb{E}_t\left[\tilde{z}^S_{t+2} - \mu_1 \tilde{z}^S_{t+1}\right] - \mu_2\left(\tilde{z}^S_{t+1} - \mu_1 \tilde{z}^S_t\right) = F_t. \tag{C.20}$$

Defining $q_t \equiv \tilde{z}^S_{t+1} - \mu_1 \tilde{z}^S_t$ gives the first-order forward equation

$$\mathbb{E}_t\left[q_{t+1}\right] - \mu_2 q_t = F_t. \tag{C.21}$$

Rearranging:

$$q_t = \mu_2^{-1}\mathbb{E}_t\left[q_{t+1}\right] - \mu_2^{-1} F_t. \tag{C.22}$$

Iterating this equation forward for $J+1$ periods gives

$$q_t = -\sum_{j=0}^{J} \mu_2^{-(j+1)}\mathbb{E}_t\left[F_{t+j}\right] + \mu_2^{-(J+1)}\mathbb{E}_t\left[q_{t+J+1}\right]. \tag{C.23}$$

The final term is the terminal assets component associated with the unstable root $\mu_2 > 1$. The forward equation alone would then allow this component to survive, generating explosive paths for liquid assets. The transversality condition must then require that

$$\lim_{J\to\infty} \mu_2^{-(J+1)}\mathbb{E}_t\left[q_{t+J+1}\right] = 0 \tag{C.24}$$

or, equivalently, that $\lim_{j\to\infty} \mu_2^{-j}\mathbb{E}_t\left[q_{t+j}\right] = 0$. Taking the limit of (C.23) then yields

$$q_t = -\sum_{j=0}^{\infty} \mu_2^{-(j+1)}\mathbb{E}_t\left[F_{t+j}\right]. \tag{C.25}$$

Using the definition of q_t, the forward solution for liquid assets is

$$\tilde{z}^S_{t+1} = \mu_1 \tilde{z}^S_t - \sum_{j=0}^{\infty} \mu_2^{-(j+1)}\mathbb{E}_t\left[F_{t+j}\right]. \tag{C.26}$$

This shows that the root μ_1 governs the persistence of bonds, while μ_2 governs their future discounting.

C.5. The Saver Consumption Function

The log-linear Saver budget constraint (C.7) can be written as

$$c^S_t = \frac{\tilde{Y}^S}{C^S}\tilde{y}^S_t + \alpha_S(\tilde{r}_t + \tilde{z}^S_t) - \alpha_B \tilde{z}^S_{t+1}. \tag{C.27}$$

The only object that is not dated t is next-period liquid assets, $\tilde{z}^S_{t+1}$. Using the forward solution (C.26) that reads

$$\tilde{z}^S_{t+1} = \mu_1 \tilde{z}^S_t - \sum_{j=0}^{\infty} \mu_2^{-(j+1)} \mathbb{E}_t \left[F_{t+j} \right],$$

the last term of the budget constraint (C.27) becomes

$$-\alpha_B \tilde{z}^S_{t+1} = -\alpha_B \mu_1 \tilde{z}^S_t + \alpha_B \sum_{j=0}^{\infty} \mu_2^{-(j+1)} \mathbb{E}_t \left[F_{t+j} \right]. \tag{C.28}$$

Substituting this expression back into the budget constraint gives

$$c^S_t = \frac{\tilde{Y}^S}{C^S} \tilde{y}^S_t + \alpha_S \tilde{r}_t + \left(\alpha_S - \alpha_B \mu_1 \right) \tilde{z}^S_t + \alpha_B \sum_{j=0}^{\infty} \mu_2^{-(j+1)} \mathbb{E}_t \left[F_{t+j} \right]. \tag{C.29}$$

To make the next step explicit, use $F_t = F^0_t / (\omega \alpha_B)$, so that $\alpha_B F_t = F^0_t / \omega$. Therefore,

$$\alpha_B \sum_{j=0}^{\infty} \mu_2^{-(j+1)} \mathbb{E}_t \left[F_{t+j} \right] = \frac{1}{\omega} \sum_{j=0}^{\infty} \mu_2^{-(j+1)} \mathbb{E}_t \left[F^0_{t+j} \right]. \tag{C.30}$$

Using the definition of F^0_t in (C.10) and the law of iterated expectations, this last term becomes

$$\begin{aligned} \frac{1}{\omega} \sum_{j=0}^{\infty} \mu_2^{-(j+1)} \Bigg\{ & \omega \frac{\tilde{Y}^S}{C^S} \mathbb{E}_t \left[\tilde{y}^S_{t+j+1} \right] + (1-\omega) \frac{\tilde{Y}^H}{C^H} \mathbb{E}_t \left[\tilde{y}^H_{t+j+1} \right] \\ & - \frac{\tilde{Y}^S}{C^S} \mathbb{E}_t \left[\tilde{y}^S_{t+j} \right] + \left[\omega \alpha_S + (1-\omega) \alpha_H - \frac{1}{\sigma} \right] \mathbb{E}_t \left[\tilde{r}_{t+j+1} \right] \\ & - \alpha_S \mathbb{E}_t \left[\tilde{r}_{t+j} \right] \Bigg\}. \end{aligned} \tag{C.31}$$

Substituting (C.31) into (C.29) gives the consumption function before collecting terms. Grouping together the variables that will enter the same line of the final expression:

$$\begin{aligned} c^S_t = & \underbrace{\frac{\tilde{Y}^S}{C^S} \tilde{y}^S_t + \frac{\tilde{Y}^S}{C^S} \sum_{j=0}^{\infty} \mu_2^{-(j+1)} \mathbb{E}_t \left[\tilde{y}^S_{t+j+1} \right] - \frac{1}{\omega} \frac{\tilde{Y}^S}{C^S} \sum_{j=0}^{\infty} \mu_2^{-(j+1)} \mathbb{E}_t \left[\tilde{y}^S_{t+j} \right]}_{\text{Saver income}} \\ & + \underbrace{\frac{1}{\omega} \sum_{j=0}^{\infty} \mu_2^{-(j+1)} \left[\omega \alpha_S + (1-\omega) \alpha_H - \frac{1}{\sigma} \right] \mathbb{E}_t \left[\tilde{r}_{t+j+1} \right] - \frac{\alpha_S}{\omega} \sum_{j=1}^{\infty} \mu_2^{-(j+1)} \mathbb{E}_t \left[\tilde{r}_{t+j} \right]}_{\text{Expected future real rates}} \\ & + \underbrace{\left(\alpha_S - \frac{\alpha_S}{\omega \mu_2} \right) \tilde{r}_t + \left(\alpha_S - \alpha_B \mu_1 \right) \tilde{z}^S_t}_{\text{Beginning-of-period wealth}} \end{aligned}$$

$$+\underbrace{\frac{1-\omega}{\omega}\frac{\tilde{Y}^H}{C^H}\sum_{j=0}^{\infty}\mu_2^{-(j+1)}\mathbb{E}_t\left[\tilde{y}^H_{t+j+1}\right]}_{\text{Future HtM income}}. \tag{C.32}$$

Equation (C.32) contains the direct budget effects and the full present-value forcing term, grouped according to the four economic blocks of the final consumption function. Rearranging it requires only shifting the indices inside the sums and simplifying the coefficients, then:

$$\begin{aligned} c_t^S \quad = \quad & \left(1-\frac{1}{\omega\mu_2}\right)\frac{\tilde{Y}^S}{C^S}\sum_{k=0}^{\infty}\mu_2^{-k}\mathbb{E}_t\left[\tilde{y}^S_{t+k}\right] \\ & + \left\{\frac{1}{\omega}\left[\omega\alpha_S+(1-\omega)\alpha_H-\frac{1}{\sigma}\right]-\frac{\alpha_S}{\omega\mu_2}\right\}\sum_{k=1}^{\infty}\mu_2^{-k}\mathbb{E}_t[\tilde{r}_{t+k}] \\ & + \alpha_S\left(1-\frac{1}{\omega\mu_2}\right)\tilde{r}_t+\left(\alpha_S-\alpha_B\mu_1\right)\tilde{z}^S_t \\ & + \frac{1-\omega}{\omega}\frac{\tilde{Y}^H}{C^H}\sum_{k=1}^{\infty}\mu_2^{-k}\mathbb{E}_t\left[\tilde{y}^H_{t+k}\right]. \end{aligned} \tag{C.33}$$

The following paragraphs explain the last rearrangements needed to obtain the final formulation of the consumption function that appears in the body of the paper.

The income block. The Saver-income line in (C.33) has the coefficient

$$1-\frac{1}{\omega\mu_2}.$$

Using the root product $\mu_1\mu_2=\beta^{-1}$, one has $\mu_2^{-1}=\beta\mu_1$. Hence

$$1-\frac{1}{\omega\mu_2}=1-\frac{\mu_1}{\omega}\beta.$$

This is the prefactor that appears in the first line of (C.36). The HtM-income block does not contain this prefactor because it is dated only from $t+1$ onward in the forcing term, and therefore enters directly as the future-income present value.

The expected future real-rate block. The coefficient on expected future real rates in (C.33) is

$$\frac{1}{\omega}\left[\omega\alpha_S+(1-\omega)\alpha_H-\frac{1}{\sigma}\right]-\frac{\alpha_S}{\omega\mu_2}.$$

Using $\omega\alpha_S+(1-\omega)\alpha_H=\alpha_B(\omega\mathcal{A}_\zeta-1)$, $\mathcal{A}_\zeta=\mu_1+\mu_2$, and $\alpha_S/\mu_2=\omega\alpha_B\mu_1$, this coefficient becomes

$$\frac{1}{\omega}\left[\alpha_B\left(\omega\mu_1+\omega\mu_2-1\right)-\frac{1}{\sigma}-\omega\alpha_B\mu_1\right]=\alpha_B\left(\mu_2-\frac{1}{\omega}\right)-\frac{1}{\omega\sigma}.$$

Finally,

$$\alpha_B\left(\mu_2-\frac{1}{\omega}\right)-\frac{1}{\omega\sigma}=\frac{\tilde{Z}^S}{C^S}\left(1-\frac{1}{\omega\mu_2}\right)\mu_2-\frac{1}{\omega\sigma}=\frac{\tilde{Z}^S}{C^S}\left(1-\frac{\mu_1}{\omega}\beta\right)\mu_2-\frac{1}{\omega\sigma}, \tag{C.34}$$

where the last equality uses $\mu_2^{-1}=\beta\mu_1$ again. This is the coefficient in the second line of (C.36).

The beginning-of-period wealth block. The beginning-of-period wealth block in (C.33) is

$$\alpha_S\left(1-\frac{1}{\omega\mu_2}\right)\tilde{r}_t+\left(\alpha_S-\alpha_B\mu_1\right)\tilde{z}_t^S.$$

Since $\alpha_S=\omega\alpha_B\mu_1\mu_2$, one has $\alpha_S/(\omega\mu_2)=\alpha_B\mu_1$. Therefore,

$$\alpha_S-\alpha_B\mu_1=\alpha_S-\frac{\alpha_S}{\omega\mu_2}=\alpha_S\left(1-\frac{1}{\omega\mu_2}\right),$$

and the current return and beginning-of-period bond stock have the same coefficient:

$$\alpha_S\left(1-\frac{1}{\omega\mu_2}\right)\left(\tilde{r}_t+\tilde{z}_t^S\right).$$

Using $\mu_2^{-1}=\beta\mu_1$ and $\alpha_S=s\tilde{R}\tilde{Z}^S/C^S$ gives

$$\left(1-\frac{\mu_1}{\omega}\beta\right)\frac{s\tilde{R}\tilde{Z}^S}{C^S}\left(\tilde{r}_t+\tilde{z}_t^S\right), \tag{C.35}$$

which is the third line of (C.36).

Rearranging (C.33) after the simplifications described in the last three paragraphs delivers that the consumption function of a Saver can be written as

$$\begin{aligned}
c_t^S \quad = \quad & \left(1-\frac{\mu_1}{\omega}\beta\right)\frac{\tilde{Y}^S}{C^S}\sum_{k=0}^{\infty}\mu_2^{-k}\mathbb{E}_t\left[\tilde{y}_{t+k}^S\right] \\
& +\left[\frac{\tilde{Z}^S}{C^S}\left(1-\frac{\mu_1}{\omega}\beta\right)\mu_2-\frac{1}{\omega\sigma}\right]\sum_{k=1}^{\infty}\mu_2^{-k}\mathbb{E}_t[\tilde{r}_{t+k}] \\
& +\left(1-\frac{\mu_1}{\omega}\beta\right)\frac{s\tilde{R}\tilde{Z}^S}{C^S}\left(\tilde{r}_t+\tilde{z}_t^S\right) \\
& +\frac{\tilde{Y}^H}{C^H}\frac{1-\omega}{\omega}\sum_{k=1}^{\infty}\mu_2^{-k}\mathbb{E}_t\left[\tilde{y}_{t+k}^H\right].
\end{aligned} \tag{C.36}$$

C.6. Intertemporal MPC Profiles

This section derives the iMPC profile out of a one-time income shock. That corresponds to the first column of the intertemporal MPC matrix in Auclert et al. (2024).

Saver iMPC profile out of own income. Assume a one-time shock to current Saver income, $\tilde{y}_0^S = 1$. At $t = 0$, the consumption function (C.36) then simplifies to

$$c_0^S = \left(1 - \frac{\mu_1}{\omega}\beta\right) \frac{\tilde{Y}^S}{C^S} \tag{C.37}$$

and from the budget constraint (C.7), the beginning-of-next-period bond position (i.e. the liquid assets carried into period $t = 1$) is

$$\tilde{z}_1^S = \frac{\tilde{Y}^S}{\tilde{Z}^S} \frac{\mu_1}{\omega}\beta. \tag{C.38}$$

From $t \geq 1$ there is no income and no rate movement, so consumption from (C.36) reduces to

$$c_t^S = \left(1 - \frac{\mu_1}{\omega}\beta\right) \frac{s\tilde{R}\tilde{Z}^S}{C^S} \tilde{z}_t^S \tag{C.39}$$

and recalling that asset persistence is governed by μ_1, one has

$$\tilde{z}_{t+1}^S = \mu_1 \tilde{z}_t^S \quad \text{therefore:} \quad \tilde{z}_t^S = \mu_1^{t-1} \tilde{z}_1^S. \tag{C.40}$$

Combining (C.39) together with (C.40), and using $s\tilde{R} = \frac{\omega}{\beta}$ from (C.2) one has

$$c_t^S \quad = \quad \left(1 - \frac{\mu_1}{\omega}\beta\right) \frac{s\tilde{R}\tilde{Z}^S}{C^S} \mu_1^{t-1} \frac{\tilde{Y}^S}{\tilde{Z}^S} \frac{\mu_1}{\omega}\beta \quad = \quad \left(1 - \frac{\mu_1}{\omega}\beta\right) \frac{\tilde{Y}^S}{C^S} \mu_1^t, \qquad t \geq 1, \tag{C.41}$$

which also reproduces (C.37) at $t = 0$. Converting to level responses, the Saver's iMPC path is then given by

$$\frac{\partial C_t^S}{\partial \tilde{Y}_0^S} \quad = \quad \frac{C^S}{\tilde{Y}^S} \frac{\partial c_t^S}{\partial \tilde{y}_0^S} \quad = \quad \left(1 - \frac{\mu_1}{\omega}\beta\right) \mu_1^t, \qquad t \geq 0. \tag{C.42}$$

On impact, the Saver consumes a fraction $\left(1 - \frac{\mu_1}{\omega}\beta\right)$ of the shock, the remaining fraction $\frac{\mu_1}{\omega}\beta$ goes into savings, through liquid bonds, which then decay at the stable root $\mu_1 \in (0, 1)$, and so does consumption. Since (C.42) involves only μ_1, ω and β, the consumption path is independent of the level of steady-state liquidity.

Aggregate iMPC profile out of aggregate income. Similarly to the previous paragraph, assume a one-time aggregate income shock with equal incidence across states, that is

$$\tilde{y}_0^S = \tilde{y}_0^H = 1.$$

The HtM income shock affects neither the Saver's consumption at $t = 0$, nor aggregate liquid assets since HtM are constrained. Hence the Saver's path is unchanged from the one described in (C.42).

The HtM on the other hand consume their income on impact, intuitively from the budget constraint (C.6):

$$\frac{\partial c_0^H}{\partial \tilde{y}_0^H} = \frac{\tilde{Y}^H}{C^H} \quad \text{and} \quad \frac{\partial C_0^H}{\partial \tilde{Y}_0^H} = 1 \tag{C.43}$$

Then, from $t \geq 1$, the HtM consume the gross payoff on the pooled bonds that are brought by transitioning Savers. From the HtM flow budget constraint (C.6) combined with bond persistence (C.40) one then has:

$$c_t^H = \frac{C^H - \tilde{Y}^H}{C^H} \mu_1^{t-1} \tilde{z}_1^S, \qquad t \geq 1. \tag{C.44}$$

Concerning aggregate responses, define aggregate consumption and income as

$$C_t = \lambda C_t^H + (1-\lambda) C_t^S \quad \text{and} \quad \tilde{Y}_t = \lambda \tilde{Y}_t^H + (1-\lambda)\tilde{Y}_t^S$$

with steady-state levels

$$C = \lambda C^H + (1-\lambda) C^S \quad \text{and} \quad \tilde{Y} = \lambda \tilde{Y}^H + (1-\lambda)\tilde{Y}^S$$

respectively. The log deviation of aggregate consumption is then the consumption-weighted average of the group log deviations, $c_t = \frac{\lambda C^H c_t^H + (1-\lambda) C^S c_t^S}{C}$, and the equal-incidence shock moves aggregate income by $\tilde{y}_0 = 1$.

On impact at $t = 0$, HtM consume the full income shock and Savers consume the fraction $1 - \frac{\mu_1}{\omega}\beta$ of their own income shock.

For $t \geq 1$, there are no other shocks. Aggregate consumption therefore comes only from the bonds accumulated by Savers at $t = 0$, and then pooled within groups after the transitions. Starting from the definition of aggregate consumption,

$$C c_t = \lambda C^H c_t^H + (1-\lambda) C^S c_t^S.$$

Using the Saver's response (C.39), the bond's persistence (C.40), and the HtM consumption path (C.44), this becomes

$$\begin{aligned} C c_t &= \lambda C^H \left[\frac{C^H - \tilde{Y}^H}{C^H} \mu_1^{t-1} \tilde{z}_1^S \right] + (1-\lambda) C^S \left[\left(1 - \frac{\mu_1}{\omega}\beta\right) \frac{s \tilde{R} \tilde{Z}^S}{C^S} \mu_1^{t-1} \tilde{z}_1^S \right] \\ &= \left[\lambda \left(C^H - \tilde{Y}^H\right) + (1-\lambda) \left(1 - \frac{\mu_1}{\omega}\beta\right) s \tilde{R} \tilde{Z}^S \right] \mu_1^{t-1} \tilde{z}_1^S. \end{aligned}$$

The first term in brackets can be rewritten using the HtM steady-state budget constraint and stationarity of population shares:

$$\lambda \left(C^H - \tilde{Y}^H\right) = \lambda (1-h) \tilde{R} \tilde{Z}^S = (1-\lambda)(1-s) \tilde{R} \tilde{Z}^S.$$

Moreover, since $s\tilde{R} = \frac{\omega}{\beta}$, then $\left(1 - \frac{\mu_1}{\omega}\beta\right) s\tilde{R} = s\tilde{R} - \mu_1$. Therefore

$$\begin{aligned} C c_t &= (1-\lambda)\left[(1-s)\tilde{R} + s\tilde{R} - \mu_1\right]\tilde{Z}^S \mu_1^{t-1}\tilde{z}_1^S \\ &= (1-\lambda)\left(\tilde{R} - \mu_1\right)\tilde{Z}^S \mu_1^{t-1}\tilde{z}_1^S. \end{aligned}$$

Finally, substituting (C.38), $\tilde{z}_1^S = \frac{\tilde{Y}^S}{\tilde{Z}^S}\frac{\mu_1}{\omega}\beta$, gives

$$c_t = (1-\lambda)\frac{\left(\tilde{R} - \mu_1\right)\tilde{Y}^S}{C}\frac{\mu_1}{\omega}\beta\,\mu_1^{t-1}, \qquad t \geq 1,$$

and therefore the aggregate consumption response in log-deviations can be summarized by:

$$c_0 = \frac{\lambda\tilde{Y}^H + (1-\lambda)\tilde{Y}^S\left(1 - \frac{\mu_1}{\omega}\beta\right)}{C}, \qquad c_t = (1-\lambda)\,\frac{\left(\tilde{R} - \mu_1\right)\tilde{Y}^S}{C}\,\frac{\mu_1}{\omega}\beta\,\mu_1^{t-1} \quad t \geq 1. \tag{C.45}$$

Converting to levels, the aggregate iMPC is the log-deviation response rescaled by the aggregate consumption-to-income ratio, $\frac{dC_t}{d\tilde{Y}_0} = \frac{C}{\tilde{Y}}\frac{c_t}{\tilde{y}_0} = \frac{C}{\tilde{Y}}c_t$. The impact aggregate response is then

$$\frac{dC_0}{d\tilde{Y}_0} = \frac{\lambda\tilde{Y}^H + (1-\lambda)\tilde{Y}^S\left(1 - \frac{\mu_1}{\omega}\beta\right)}{\lambda\tilde{Y}^H + (1-\lambda)\tilde{Y}^S}, \tag{C.46}$$

and for $t \geq 1$

$$\frac{dC_t}{d\tilde{Y}_0} = (1-\lambda)\,\frac{\left(\tilde{R} - \mu_1\right)\tilde{Y}^S}{\lambda\tilde{Y}^H + (1-\lambda)\tilde{Y}^S}\,\frac{\mu_1}{\omega}\beta\,\mu_1^{t-1}. \tag{C.47}$$

C.7. Comparison with the Representative-Agent Benchmark

The representative-agent (RA) benchmark is nested at $\omega = 1$: there $\mathcal{A}_\zeta = 1 + \beta^{-1}$, so the characteristic polynomial factors as $P(\mu) = (\mu - 1)\left(\mu - \beta^{-1}\right)$ and

$$\mu_1\big|_{\omega=1} = 1, \qquad \mu_2\big|_{\omega=1} = \beta^{-1}: \tag{C.48}$$

the impact MPC is $1 - \beta$, future income is discounted at β, and the response to the k-period-ahead rate is $-\frac{1}{\sigma}\beta^k$. This section compares the THANK responses with this benchmark, term by term. All three comparisons reduce to the position of the polynomial $P(\mu) = \mu^2 - \mathcal{A}_\zeta\mu + \beta^{-1}$ at a single evaluation point, since $P(\mu) < 0$ exactly for $\mu \in (\mu_1, \mu_2)$.

The impact MPC. The Saver's impact MPC exceeds the RA value if and only if

$$1 - \frac{\mu_1}{\omega}\beta \;>\; 1 - \beta \quad \Longleftrightarrow \quad \mu_1 < \omega \quad \Longleftrightarrow \quad P(\omega) < 0. \tag{C.49}$$

Substituting (C.12) and factoring:

$$P(\omega) = \omega^2 - \mathcal{A}_\zeta\,\omega + \beta^{-1} = (1-\omega)\left[\frac{(1+\omega)(1-\beta)}{\beta} - \frac{\omega}{\beta}\,\frac{(1-h)\,\zeta}{s}\right], \tag{C.50}$$

so that, with $1-\omega>0$ and $1+\frac{1}{\omega}=\frac{2s+(1-s)\zeta^{\sigma}}{s}$,

$$\mu_1<\omega \quad\Longleftrightarrow\quad (1-h)\,\zeta \;>\; (1-\beta)\left[2s+(1-s)\,\zeta^{\sigma}\right]. \tag{C.51}$$

This is the two-period condition for front-loading to prevail over self-insurance (Appendix B.8.6), with the hurdle scaled by $1-\beta$: extending the horizon from two periods to infinity shrinks the threshold by the factor $1-\beta$. At quarterly discount factors the right-hand side of (C.51) is of order 10^{-2}, so the condition holds except in a negligible neighbourhood of $h=1$: in the infinite-horizon model, front-loading generally prevails and the Saver's impact MPC exceeds the RA value.

The profile of the response to future income. From (C.36), the Saver's responses to income at different horizons satisfy

$$\frac{M_{0,k}}{M_{0,0}}=\mu_2^{-k}=\left(\beta\mu_1\right)^k, \qquad M_{0,k}\equiv\frac{\partial C_0^S}{\partial\,\mathbb{E}_0[\tilde{Y}_k^S]}, \tag{C.52}$$

against β^k in the RA benchmark. The profile tilts towards the present — future income is discounted more steeply than by β — if and only if $\mu_1<1$, which is the saddle condition (C.19) itself. In contrast with the two-period model, where the tilt requires $\mathcal{F}>1$ and fails at standard calibrations, at the infinite horizon the tilt holds whenever the model has a solution: the per-period wedge of the two-period model, $\frac{1}{\mathcal{F}}=\frac{\phi}{\omega}$, which may exceed one, is replaced by its compounded counterpart μ_1, which cannot.

Intertemporal substitution. The response to the k-period-ahead expected rate is $-\frac{1}{\sigma\omega}\left(\beta\mu_1\right)^k$ (abstracting from the wealth-income terms, which are scaled by $\frac{\tilde{Z}^S}{C^S}$ and vanish at zero liquidity), against $-\frac{1}{\sigma}\beta^k$ in the RA benchmark. The ratio of the two is

$$\frac{\mu_1^k}{\omega}, \tag{C.53}$$

so the substitution response is attenuated at every horizon $k\geq 1$ whenever $\mu_1<\omega$ — the same condition (C.51) that amplifies the impact MPC.

C.8. Relation to Bilbiie (2025) and to Auclert et al. (2024)

Bilbiie (2025) derives analytical iMPCs for the THANK model at the *symmetric* zero-liquidity steady state ($C^H=C^S$, $\tilde{R}=\beta^{-1}$): his equation (31) is a second-order difference equation in aggregate liquid assets with trace $\Theta\equiv\frac{1}{s}+\beta^{-1}\left[1+\frac{1-s}{s}\left(\frac{1-s}{\lambda}-1\right)\right]$, and his Proposition 8 gives the resulting iMPC profile, decaying geometrically at the stable root x_b. This appendix nests that result. Setting $\zeta=1$ (so $\omega=s$) in (C.12) and using the stationarity of the shares, a direct computation gives

$$\mathcal{A}_{\zeta=1}=\frac{1}{s}+\beta^{-1}\left[1+\frac{1-s}{s}\left(\frac{1-s}{\lambda}-1\right)\right]=\Theta, \tag{C.54}$$

so that $\mu_1|_{\zeta=1} = x_b$ and the iMPCs of Section C.6 reproduce his Proposition 8 at the symmetric steady state.

Relative to that benchmark, the derivation here generalizes along three dimensions. First, the steady state is unequal: $\zeta > 1$ replaces the transition probability s by the marginal-utility weight $\omega < s$ in the asset dynamics, and depresses the steady-state return, $\tilde{R} < \beta^{-1}$. Since $\mathcal{A}_\zeta$ moves with ζ and σ, steady-state inequality is a determinant of iMPC *persistence* — a margin absent in the symmetric case, where the stable root depends on (λ, s, β) only; cyclical income inequality, the margin emphasized by Bilbiie (2025), raises the impact response but leaves x_b unchanged. Second, liquidity is positive: the consumption function (C.36) evaluates the full path of real rates (the revaluation of the outstanding stock and the wealth-income effect of expected future rates) channels that vanish at zero liquidity. Third, the object derived is the household-level consumption function in arbitrary income and rate paths, before any equilibrium income closure; the state-by-state iMPCs of Section C.6 accordingly sharpen the aggregate profile: the kink between the impact response and the decay phase originates entirely in the HtM spike, while the Saver's own path is geometric from impact.

This last point admits a useful reformulation in the language of the sequence-space literature. The consumption function (C.36) expresses c_t^S as a linear functional of the expected sequences $\{\mathbb{E}_t[\tilde{y}^S_{t+k}]\}$, $\{\mathbb{E}_t[\tilde{y}^H_{t+k}]\}$ and $\{\mathbb{E}_t[\tilde{r}_{t+k}]\}$ and of the bond position; combined with the law of motion (C.26), which solves the position forward in the same sequences. It then delivers the consumption *sequence* as a linear map of the shock sequences and the initial condition. The consumption functions therefore constitute the THANK household block in *sequence space*: its intertemporal Jacobians. The matrices of partial derivatives of consumption at every date with respect to income and rates at every date, which Auclert et al. (2024) compute numerically for quantitative heterogeneous-agent models, are here available in closed form

Appendix D. Cohorts Decomposition in THANK

This appendix confronts the state-contingent consumption functions of Appendix C with the quantitative two-asset heterogeneous-agent benchmark (HA-two) of Auclert et al. (2024), henceforth ARS

D.1. Deriving Fixed-Cohort Consumption in THANK

The comparison below uses fixed cohorts. This means that households are grouped according to their state at the date of the shock, $t = 0$, and are then followed over time.

Let $\mathrm{d}c_t^H$ and $\mathrm{d}c_t^S$ denote the per-capita state-level consumption responses implied by the consumption functions of Appendix C. These are current-state objects: $\mathrm{d}c_t^H$ is the response of a household who is HtM at date t, while $\mathrm{d}c_t^S$ is the response of a household who is a Saver at date t. Because THANK features pooling within states, current consumption depends on the current state, not on the household's history.

Consider first a household who is HtM at $t = 0$. The probability that this household is HtM at date $t + 1$ is

$$\Pr(H_{t+1} \mid H_0) = h \Pr(H_t \mid H_0) + (1 - s)[1 - \Pr(H_t \mid H_0)]. \tag{D.1}$$

The first term is the probability that the household is HtM at t and remains HtM; the second is the probability that the household is a Saver at t and transitions into the HtM state. Rearranging gives

$$\Pr(H_{t+1} \mid H_0) = (1 - s) + \rho \Pr(H_t \mid H_0), \qquad \rho \equiv s + h - 1. \tag{D.2}$$

The stationary HtM share λ satisfies the same equation,

$$\lambda = (1 - s) + \rho\lambda, \tag{D.3}$$

which is equivalent to the stationarity identity $(1 - \lambda)(1 - s) = \lambda(1 - h)$. Subtracting (D.3) from (D.2) yields

$$\Pr(H_{t+1} \mid H_0) - \lambda = \rho[\Pr(H_t \mid H_0) - \lambda]. \tag{D.4}$$

Define the distance from the stationary HtM share as $x_t \equiv \Pr(H_t \mid H_0) - \lambda$. Then (D.4) becomes $x_{t+1} = \rho x_t$, so $x_t = \rho^t x_0$. Since the household starts in the HtM state, $\Pr(H_0 \mid H_0) = 1$, and therefore $x_0 = 1 - \lambda$. Substituting this initial condition into $x_t = \rho^t x_0$ gives

$$\eta_t^{H_0} \equiv \Pr(H_t \mid H_0) = \lambda + (1 - \lambda)\rho^t. \tag{D.5}$$

The same argument applies to a household who starts as a Saver. The only difference is the initial condition, $\Pr(H_0 \mid S_0) = 0$. Therefore,

$$\eta_t^{S_0} \equiv \Pr(H_t \mid S_0) = \lambda\left(1 - \rho^t\right). \tag{D.6}$$

The fixed-cohort consumption responses now follow by averaging the state-level responses using these occupancy probabilities. For the cohort that starts HtM, a fraction $\eta_t^{H_0}$ is HtM at date t and consumes according to $\mathrm{d}c_t^H$, while a fraction $1 - \eta_t^{H_0}$ is Saver and consumes according to $\mathrm{d}c_t^S$. Since the initial HtM cohort has mass λ, its contribution to the aggregate iMPC is

$$C_t^{H_0} = \lambda\left[\eta_t^{H_0}\,\mathrm{d}c_t^H + \left(1 - \eta_t^{H_0}\right)\mathrm{d}c_t^S\right]. \tag{D.7}$$

Similarly, for the cohort that starts as Saver, a fraction $\eta_t^{S_0}$ is HtM at date t and a fraction $1 - \eta_t^{S_0}$ is Saver. Since the initial Saver cohort has mass $1 - \lambda$, its contribution is

$$C_t^{S_0} = (1 - \lambda)\left[\eta_t^{S_0}\,\mathrm{d}c_t^H + \left(1 - \eta_t^{S_0}\right)\mathrm{d}c_t^S\right]. \tag{D.8}$$

Combining (D.7) and (D.8), the cohort contributions can be written compactly as

$$C_t^{H_0} = \lambda \left[\eta_t^{H_0} \, \mathrm{d}c_t^H + \left(1 - \eta_t^{H_0}\right) \mathrm{d}c_t^S \right], \qquad C_t^{S_0} = (1-\lambda) \left[\eta_t^{S_0} \, \mathrm{d}c_t^H + \left(1 - \eta_t^{S_0}\right) \mathrm{d}c_t^S \right], \tag{D.9}$$

with

$$\eta_t^{H_0} = \lambda + (1-\lambda)\rho^t, \qquad \eta_t^{S_0} = \lambda \left(1 - \rho^t\right). \tag{D.10}$$

D.2. HANK and THANK Calibration

Table 2: Calibration of HANK and THANK used in Figure 1.

	HANK	THANK
Model	HA-two	THANK
Source	Auclert et al., 2024	Bilbiie, 2025
Solution	Sequence-space Jacobian	Analytical
Discount factor	$\beta = 0.9309$	$\beta = 0.95$
Real interest rate	$r = 0.05$	$\beta^{-1} - 1 = 0.0526$
IES	$1/\sigma = 1$	$1/\sigma = 1$
Income process persistence	$\rho_e = 0.91$	–
Income shock s.d.	$\sigma_e = (1-\theta)0.92$	–
Tax progressivity	$\theta = 0.181$	–
Liquid asset grid	$b \in [0, 10000]$, $n_b = 50$	–
Illiquid asset grid	$a \in [0, 10000]$, $n_a = 50$	–
Productivity grid	$n_e = 11$	–
Liquid return spread	$\zeta^{ARS} = 0.08/(1+r)$	–
Calvo adjustment probability	$\nu = 0.0891$	–
HtM share	–	$\lambda = 0.33$
Saver persistence	–	$s = 0.82$
HtM persistence	–	$h = 0.6345$
Consumption ratio	–	$\zeta = C^S/C^H = 1$
HtM shock loading	–	$\chi_H = 1.4$
Saver shock loading	–	induced by aggregation
Constrained definition	$b_0 \leq 0.0498$	H_0 state
Constrained cohort mass	0.7539	0.33
Constrained cohort income share	0.5920	–